

The Laguerre-Pólya Class, Jensen Polynomials and the Riemann Hypothesis

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Abstract

This thesis is primarily an exposition on an approach to the Riemann Hypothesis that was first proposed by Pólya [11] in 1927. This approach exploits the equivalence of the Riemann Hypothesis to the hyperbolicity¹ of the *Jensen polynomials*

$$J^d(x) = \sum_{k=0}^d \binom{d}{k} \gamma(k) z^k \tag{1}$$

for each integer $d \geq 1$, where $\gamma(k)$ is the coefficient of z^k in the power series for $\xi(1/2 + z)$.

At the heart of this equivalence is Pólya and Schur's [12] remarkable characterisation theorem for (what is now known as) the Laguerre-Pólya class of functions. This class consists of entire functions that can be uniformly approximated on compact sets by hyperbolic polynomials. We give the (to the author's knowledge) first English translation of [12] and summarise and outline the proofs of its main theorems.

Interest in this line of attack was reinvigorated in 2019 when Griffin, Ono, Rolin and Zagier ([6], [7]) proved the hyperbolicity of 'many' of these Jensen polynomials. In particular, they showed that for each $d \geq 1$, the shifted Jensen polynomial

$$J^{d,n}(z) = \sum_{k=0}^d \binom{d}{k} \gamma(k+n) z^k$$

is hyperbolic for all sufficiently large n . This result is based on an asymptotic expansion for the coefficients of the Riemann xi function. We present O'Sullivan's [9] proof of this asymptotic expansion.

O'Sullivan, in the same paper, introduced another family of modified Jensen polynomials whose hyperbolicity is equivalent to the Riemann hypothesis. We give a proof of the hyperbolicity of these modified polynomials, which is based on the same asymptotic expansion.

¹A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is said to be hyperbolic if all its zeroes are real.

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Contents

1	The Zeta Function and Riemann Hypothesis	8
1.1	Infinite products	8
1.2	Order and genus of an entire function	10
1.3	Gamma function	11
1.4	Zeta function	15
1.5	Jacobi theta function	17
1.6	Functional equation for the zeta function	19
1.7	Two properties of the Riemann xi function	20
2	A Polynomial Miscellany	23
2.1	Basic theory and location of zeroes	23
2.2	Hermite Polynomials	27
2.3	Sturm sequences	31
2.4	Schur's composition theorem	33
3	The Laguerre-Pólya Class	38
3.1	Multiplier sequences	38
3.2	Jensen polynomials	40
3.3	Pólya and Schur's characterisation theorems	41
3.4	Applications to the Riemann Hypothesis	45
4	Asymptotics for the Riemann xi function	49
4.1	The integral form of $\xi^{(2n)}(1/2)$ revisited	49
4.2	Preliminaries for O'Sullivan's expansion	50
4.2.1	Lambert W function	50
4.2.2	The function $h_u(x)$	52
4.2.3	Bounds for the incomplete gamma function	55
4.3	Suitable functions	56
4.4	O'Sullivan's expansion	57

4.5	Asymptotic expansion of $\xi^{(2n)}(1/2)$	62
5	Hyperbolicity of modified Jensen polynomials	64
5.1	Jensen polynomials	65
5.2	O'Sullivan's modified Jensen polynomials	65

Introduction

In 1859, Riemann published a landmark paper [16] that became the foundation of analytic number theory. Though only six pages long, it introduced the study of the zeta function as a meromorphic function. Prior to this, the zeta function had only been examined for real values—for instance, by Euler, who determined $\zeta(2) = \frac{\pi^2}{6}$ (known as the Basel problem). Riemann was the first to investigate the zeta function as a function of a complex variable.

In this paper, Riemann conjectured that all non-trivial zeros of the zeta function have real part equal to $1/2$. Though he believed this to be almost certainly true, after some ‘fleeting vain attempts’ (his own words) at a proof, he chose to proceed with other investigations. This conjecture is now known as *the Riemann hypothesis*.

More than 150 years later, despite many attempts, the Riemann hypothesis remains open. One such attempt was suggested by Pólya in 1927. He hoped to exploit the equivalence of the Riemann hypothesis to the hyperbolicity of a sequence of polynomials.

In this thesis, we prove this equivalence (along with another equivalence in terms of O’Sullivan’s modified Jensen polynomials). We also summarize recent progress in establishing the hyperbolicity of these two Jensen families.

Chapter 1 begins with a brief overview of infinite products and Hadamard theory, which are essential for Chapter 3 (on the Laguerre-Pólya class of functions). We define the order and genus of an entire function and compile key results for the reader’s convenience. Toward the end of the chapter, we prove that the Riemann xi function has order 1.

The primary objective of Chapter 1 is to state the Riemann hypothesis precisely. To this end, we analytically continue the zeta function $\zeta(s)$, starting from its Dirichlet series on $\operatorname{Re} s > 1$, to a meromorphic function with a simple pole at $s = 1$. This is achieved via the functional equation for the Riemann xi function. Our derivation of the functional equation follows one of the two proofs given by Riemann in his original 1859 paper.

Chapter 2 first covers the basic theory of orthogonal polynomials. We apply this theory to analyze the zeros of Hermite polynomials (one of the three classical orthogonal polynomial families). The discussion concludes with a lemma by Turán [20] on the zeros of a linear combination of Hermite polynomials. The second part of Chapter 2 provides a complete proof of Schur’s composition theorem, originally presented in [17]. Our exposition closely

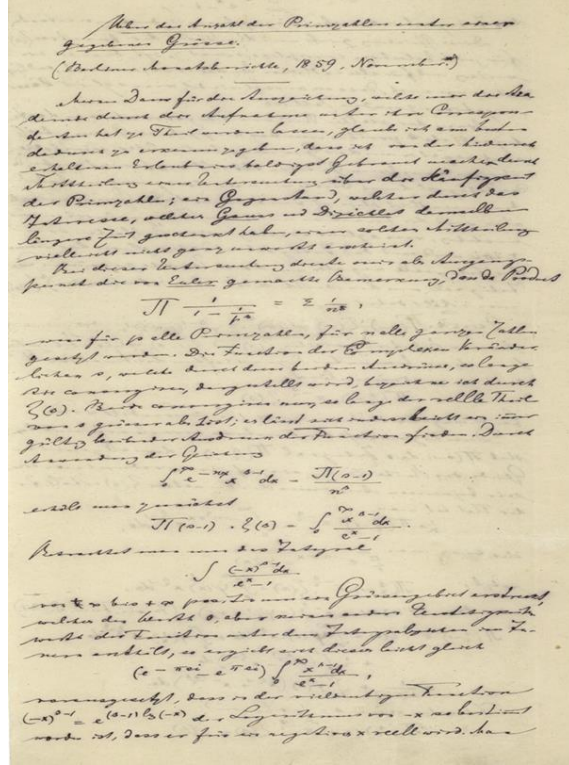


Figure 1: Riemann and the first page of his 1859 manuscript.

follows Schur's original work.

Chapter 3 focuses on the Laguerre-Pólya class of entire functions, which consists of real entire functions that can be uniformly approximated on compact sets by polynomials with only real roots. Those functions in the Laguerre-Pólya class whose approximating polynomials have roots of the same sign are called type I functions. In this chapter, we present and prove Pólya and Schur's [12] striking characterization of both the type I subclass and the full Laguerre-Pólya class. Their proof relies on two key components: Schur's composition theorem and Hadamard theory.

Chapter 4 examines O'Sullivan's [9] main result on the asymptotic expansion of certain "suitable" functions. We then apply this method to derive an asymptotic expansion for the coefficients of the Riemann xi function.

Chapter 5 establishes the equivalence of the Riemann Hypothesis and the hyperbolicity of the Jensen and modified Jensen polynomials. We also prove O'Sullivan's [9] result establishing the hyperbolicity of the modified Jensen polynomials for sufficiently large integers d and n .

Chapter 6 presents the first English translation, to the author's knowledge, of Pólya and Schur's [12] paper on the Laguerre-Pólya class. The paper [12], along with Schur's earlier work [17], was written in German and published together in issue 144 of *Crelle's Journal* in

1914. Though technically separate, these two papers are closely linked, as the joint work with Pólya extensively cites Schur's results. Due to time constraints, Schur's paper has not been translated, but key results necessary for understanding [12] are presented in Chapter 2 (particularly in the final section) of this thesis.

Chapter 1

The Zeta Function and Riemann Hypothesis

In this chapter, we take as our starting point the zeta function on reals greater than 1 and analytically continue it via its functional equation. We also precisely state the RH, and cover some essential facts about the gamma function and the basic Jacobi theta function that will be needed along the way. Our treatment follows the first chapter of Edwards' book on the zeta function [4], which offers an insightful commentary on Riemann's 1859 paper.

1.1 Infinite products

We fix the principal branch of the logarithm. Let $\{p_n\}_{n=1}^{\infty}$ be a sequence of complex numbers. We denote the n -th partial product of this sequence by

$$P_n = \prod_{k=1}^n p_k = p_1 p_2 \cdots p_n.$$

We say that the infinite product

$$\prod_{k=1}^{\infty} p_k$$

converges to P if $P_n \rightarrow P$. If an infinite product converges to a number other than zero, then we must have

$$\lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} \frac{P_n}{P_{n-1}} = 1.$$

Conversely, if $p_n \rightarrow 1$, then there is an n_0 such that $p_n \in \mathbb{C} \setminus (-\infty, 0]$ for all $n > n_0$. Suppose further that $\sum_{k=n_0+1}^{\infty} \log p_k$ converges. Then

$$\lim_{n \rightarrow \infty} P_n = p_1 \cdots p_{n_0} \lim_{n \rightarrow \infty} \exp \left(\sum_{k=n_0+1}^n \log p_k \right) = p_1 \cdots p_{n_0} \exp \left(\sum_{k=n_0+1}^{\infty} \log p_k \right).$$

That is the infinite product converges to a non-zero number, provided that none of the terms of the product are zero.

Proposition 1.1. *Let $p_n = 1 + a_n$ ($n \in \mathbb{N}$) be a sequence of non-zero complex numbers. The series $\sum_{n=1}^{\infty} a_n$ converges absolutely if and only if $\sum_{n=1}^{\infty} \log p_n$ converges absolutely¹.*

Proof. Since

$$\lim_{z \rightarrow 0} \frac{\log(1+z)}{z} = 1,$$

for every $\varepsilon > 0$, we have for all z sufficiently close to 0

$$(1 - \varepsilon)|z| < |\log(1+z)| < (1 + \varepsilon)|z|,$$

which proves both directions of the proposition. $\square$

We obtain a simple test for the convergence of an infinite product: $\prod_{n=1}^{\infty} (1 + a_n)$ (with $a_n \neq -1$ for any n) converges if the series $\sum_{n=1}^{\infty} |a_n|$ converges.

We now consider infinite products whose terms are functions of a complex variable. A natural question: if each term is analytic, and the infinite product converges in a region, is the infinite product analytic in that region? We give a partial answer in the form of the next proposition which is similar to (and in fact, uses in its proof) the Weierstrass M -test for infinite series.

Proposition 1.2. *Let $\Omega \subset \mathbb{C}$ be a region and $f_n : \Omega \rightarrow \mathbb{C}$ ($n \in \mathbb{N}$) be a sequence of functions. Suppose there exist constants M_n ($n \in \mathbb{N}$) such that $\sum_{n=0}^{\infty} M_n < \infty$ and $|f_n(z)| \leq M_n$ for each n and all $z \in \Omega$. Then the partial products $F_N(z) := \prod_{n=1}^N (1 + f_n(z))$ converge uniformly to the infinite product $\prod_{n=1}^{\infty} (1 + f_n(z))$ on Ω .*

Proof. For each $z \in \Omega$, $F(z) = \prod_{n=1}^{\infty} (1 + f_n(z))$ converges since $\sum_{n=1}^{\infty} |f_n(z)|$ does.

We choose a constant C such that $|\log(1+z)| \leq C|z|$ for $|z| \leq \frac{1}{2}$ and N such that $M_n \leq \frac{1}{2}$ for $n \geq N$. Then we have that

$$|\log(1 + f_n(z))| \leq CM_n$$

for $|z| \leq \frac{1}{2}$ and $n \geq N$. Weierstrass M -test implies that, as $m \rightarrow \infty$, $\sum_{n=N}^m \log(1 + f_n(z))$ converges uniformly on Ω , and thus so does $\sum_{n=N}^m (1 + f_n(z))$.

Since each $f_n(z)$ is bounded, on multiplication by the first $N - 1$ terms of the infinite product, we get that $\prod_{n=1}^m (1 + f_n(z))$ converges uniformly to the infinite product on Ω . $\square$

Corollary 1.0.1. *Assume along with the conditions of Proposition 1.2 that each f_n is analytic. Then $\prod_{n=1}^{\infty} (1 + f_n(z))$ is analytic in z on Ω .*

Proof. Each partial product is an analytic function, which converges uniformly on compact subsets of Ω to the infinite product. $\square$

¹If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$, and hence $p_n \rightarrow 1$. So except for finitely many n , $\log p_n$ is defined.

1.2 Order and genus of an entire function

We define the *canonical factors*

$$E_0(z) = 1 - z$$

and

$$E_k(z) = (1 - z) \exp \left(z + \frac{z^2}{2} + \cdots + \frac{z^k}{k} \right)$$

for each integer $k > 0$.

Let $\{a_n\}_{n \in \mathbb{N}}$ be a sequence of non-zero complex numbers such that $|a_n| \rightarrow \infty$ as $n \rightarrow \infty$. Then the infinite product

$$\prod_{n=1}^{\infty} E_{m_n}(z/a_n), \quad (1.1)$$

where $m_n \in \mathbb{N}$, converges to an entire function provided that

$$\sum_{n=1}^{\infty} \frac{1}{m_n + 1} \left(\frac{R}{|a_n|} \right)^{m_n + 1} \quad (1.2)$$

converges for every $R > 0$. This is the case if we take $m_n = n$ (since $|a_n| \rightarrow \infty$.)

Let f be an entire function with a zero of order $m \geq 0$ at the origin, and whose other zeroes consist precisely of $\{a_n\}$. Either $|a_n| \rightarrow \infty$ as $n \rightarrow \infty$, or the zeroes of f have a limit point and hence f is identically equal to zero. In the former case, since

$$\frac{f(z)}{z^m \prod_{n=1}^{\infty} E_n(z/a_n)}$$

is an entire function which does not vanish at any point of the complex plane, it is equal to $e^{g(z)}$ for some entire function g . Thus, we have that

$$f(z) = z^m e^{g(z)} \prod_{n=1}^{\infty} E_{m_n}(z/a_n). \quad (1.3)$$

Suppose there exists an integer $h \geq 0$ such that (1.1) converges for $m_n = h$. Then (1.3) takes the form

$$f(z) = z^m e^{g(z)} \prod_{h=1}^{\infty} E_h(z/a_n) \quad (1.4)$$

for some entire function g .

Definition 1.1. We call (1.4) the canonical product for f if h is the smallest possible integer such that (1.2) converges (for each $R > 0$) with $m_n = h$. That is, if h is the smallest integer such that

$$\sum_{n=1}^{\infty} \frac{1}{|a_n|^{h+1}} < \infty.$$

If in the canonical product for f the function g is a polynomial, then we say that f has finite genus equal to $\max(h, \deg g)$.

Hadamard's theorem, which we will state precisely shortly, says that a function for which $|f(z)|$ does not grow 'too fast' as $|z| \rightarrow \infty$ has finite genus. Conversely, functions with finite genus exhibit this kind of tame growth.

Definition 1.2. The entire function $f(z)$ is said to be of order $\leq \rho$ if there exist constants $A, B > 0$ such that

$$|f(z)| \leq Ae^{B|z|^\rho} \quad (1.5)$$

for all $z \in \mathbb{C}$. We define the order of f as the infimum of the set of all $\rho > 0$ such that f is of order $\leq \rho$.

Theorem 1.1 (Hadamard's factorization theorem). *The genus h and order ρ of a function f satisfy the double inequality*

$$h \leq \rho < h + 1.$$

In particular, they are either both finite or both infinite.

A proof of this theorem (as well as a proof of the convergence criterion (1.2)) may be found in Chapter 5 of [1].

1.3 Gamma function

The factorial function $n!$ is defined for $n \in \mathbb{N}$. It can be continued, as we will see in the next paragraph, on complex numbers with real part greater than zero using the Euler integral. This latter function, which is called the gamma function, can be analytically continued to a meromorphic function with simple poles at the negative integers. We have consulted Freitag and Busam's textbook [5] and Remmert's expository note on Wielandt's theorem [15] in this section.

Let σ and t denote respectively the real and imaginary parts of $s \in \mathbb{C}$. Let $\mathbb{C}_+$ denote the half-plane given by $\sigma > 0$. For $s \in \mathbb{C}_+$, we define the *gamma function* as

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt. \quad (1.6)$$

A simple integration by parts shows that $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}^*$. The right-hand side of (1.6) is finite for $\sigma > 0$: if we pick L such that $t^{\sigma-1} < e^{t/2}$ for all $t \geq L$, then

$$\begin{aligned} |\Gamma(s)| &\leq \int_0^\infty |t^{s-1} e^{-t}| dt = \int_0^\infty t^{\sigma-1} e^{-t} dt \\ &= \int_0^1 t^{\sigma-1} e^{-t} dt + \int_1^\infty t^{\sigma-1} e^{-t} dt \\ &\leq \int_0^1 t^{\sigma-1} dt + \int_1^L t^{\sigma-1} e^{-t} dt + \int_L^\infty e^{-t/2} dt \end{aligned}$$

which is finite for $\sigma > 0$.

It follows easily from Morera's theorem that $\Gamma(s)$ is analytic on $\mathbb{C}_+$.

Proposition 1.3. *The function defined by (1.6) has the following properties:*

- (a) $\Gamma(1) = 1$.
- (b) $\Gamma(s+1) = s\Gamma(s)$ for $s \in \mathbb{C}_+$. In particular, $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$.
- (c) $|\Gamma(s)|$ is bounded on the vertical strip given by $1 \leq \sigma \leq 2$.

Proof. Part (a) follows by plugging $s = 1$ in (1.6). Integrating by parts, we get

$$\begin{aligned}\Gamma(s+1) &= \int_0^\infty t^s e^{-t} dt = -t^s e^{-t} \Big|_0^\infty + s \int_0^\infty t^{s-1} e^{-t} dt \\ &= s\Gamma(s).\end{aligned}$$

Repeated application of this identity combined with part (a) proves (b). Since Γ is analytic, $\Gamma(\sigma)$ attains its maximum value on the interval $[1, 2]$. Since $|\Gamma(s)| \leq \Gamma(\sigma)$, we have that $|\Gamma(s)|$ is bounded on the strip $1 \leq \sigma \leq 2$. □

The equation stated in part (b) of the above proposition is called the *functional equation* of the gamma function. Wielandt's Theorem, a proof of which may be found in [15], says that if f is an analytic function on $\mathbb{C}_+$ which satisfies the functional equation of the gamma function and is bounded on the vertical strip given by $1 \leq \sigma \leq 2$, then f must be a constant multiple of Γ . Additionally, this constant is given by $f(1)$. Therefore, we have

Theorem 1.2. *$\Gamma(s)$ is the only analytic function on $\mathbb{C}_+$ which satisfies the three conditions of Proposition 1.3.*

Let s denote a complex number other than a negative integer or zero. We define the *Gauss product* as

$$G(s) = \lim_{n \rightarrow \infty} \frac{n! n^{s-1}}{s(s+1) \dots (s+n-1)}. \quad (1.7)$$

We picked s so that it would not be a root of the denominator of (1.7). In what follows, we show that G is the analytic continuation of Γ .

G is analytic on $\mathbb{C} \setminus \{0, -1, -2, \dots\}$. Telescoping allows us to write

$$\begin{aligned}\frac{n! n^{s-1}}{(s+1) \dots (s+n-1)} &= \frac{n^s}{s} \prod_{k=1}^{n-1} \left(\frac{1}{1 + \frac{s}{k}} \right) \\ &= \frac{1}{s} \prod_{k=1}^{n-1} \left(1 + \frac{s}{k} \right)^{-1} \left(1 + \frac{1}{k} \right)^s.\end{aligned}$$

Since

$$\left(1 + \frac{s}{k}\right)^{-1} \left(1 + \frac{1}{k}\right)^s = 1 + \frac{s(s-1)}{k^2} + O\left(\frac{1}{k^3}\right),$$

the infinite product converges, and analyticity follows from the fact that each partial product is analytic.

G has simple poles at the non-positive integers. It follows easily from (1.7) that $G(1) = 1$ and $G(s+1) = sG(s)$, which implies, for non-positive integer n ,

$$\lim_{s \rightarrow -n} (s+n)G(s) = \lim_{s \rightarrow -n} \frac{G(s+n+1)}{s(s+1)\dots(s+n-1)} = \frac{(-1)^n}{n!}.$$

$G(s)$ does not vanish for any s . The infinite product in (1.7), whose terms never vanish for any valid s , converges to a non-zero number.

$G(s)$ is bounded on $1 \leq \sigma \leq 2$. Since $|s+k| \geq \sigma+k$ ($k = 0, 1, \dots, n-1$) and $|n^{s-1}| \leq n^{\sigma-1}$, we have that

$$|G(s)| = \lim_{s \rightarrow \infty} \left| \frac{n!n^{s-1}}{s(s+1)\dots(s+n-1)} \right| \leq \lim_{n \rightarrow \infty} \frac{n!n^{\sigma-1}}{\sigma(\sigma+1)\dots(\sigma+n-1)} = G(\sigma),$$

which is bounded for $\sigma \in [0, 1]$.

The conditions of Theorem 1.2 hold. Thus, $G|_{\mathbb{C}_+} = \Gamma$. We simplify our notation and use $\Gamma(s)$ in the sense of its analytic continuation $G(s)$. We record this discussion as the following theorem.

Theorem 1.3. Γ is a meromorphic function with simple poles at the non-positive integers which satisfies the functional equation $\Gamma(s+1) = s\Gamma(s)$. Moreover, $\Gamma(1) = 1$ and $\Gamma(s)$ does not vanish for any $s \in \mathbb{C} \setminus \{0, -1, -2, \dots\}$.

We denote the n -th harmonic number by $H_n := \sum_{k=1}^n 1/k$. As $n \rightarrow \infty$, $H_n - \log(n)$ converges to a constant $\gamma \approx 0.577\dots$, which is known as the Euler–Mascheroni constant. We will exploit this fact to prove the next theorem.

Theorem 1.4. The reciprocal of the gamma function is entire and has the Weierstrass product

$$\frac{1}{\Gamma(s)} = se^{\gamma s} \prod_{k=1}^{\infty} \left(1 + \frac{s}{k}\right) e^{-s/k} \quad (1.8)$$

where γ is the Euler–Mascheroni constant.

Proof. The fact that $1/\Gamma$ is entire is evident from Theorem 1.3. The infinite product on the right of (1.8) converges as

$$\left(1 + \frac{s}{k}\right) e^{-s/k} = 1 - \frac{s^2}{2k^2} + O\left(\frac{1}{k^3}\right).$$

The n -th term of the reciprocal Gauss product can be written as

$$\begin{aligned}\frac{s(s+1)\dots(s+n-1)}{n!n^{s-1}} &= \frac{s(s+1)\dots(s+n)}{n!n^s} \\ &= e^{-s\log(n)} s \left(1 + \frac{s}{1}\right) \dots \left(1 + \frac{s}{n}\right) \\ &= s e^{(H_n - \log(n))s} \prod_{k=1}^{\infty} \left(1 + \frac{s}{k}\right) e^{-s/k}.\end{aligned}$$

We apply the limit $n \rightarrow \infty$ on both sides and prove (1.8). □

Stirling's approximation provides an asymptotic estimate for the behaviour of $\Gamma(s)$ as $|s| \rightarrow \infty$. One way of proving Stirling's approximation is using the Euler-Maclaurin summation formula (see chapter three of de Bruijn's book on asymptotic analysis [3]). However since we only require the approximation for real s , we use *Laplace's method*, which is a technique for obtaining asymptotics for real integrals that depend on a (real) parameter. This method will be also be used to prove asymptotics for the Riemann xi function in Chapter 4.

The following theorem is proved in §4.2 of [3].

Theorem 1.5 (Laplace's method). *Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function which satisfies the following conditions.*

- (a) $h(t)$ has an absolute maximum at $t = 0$.
- (b) There exist positive reals b and c such that $h(t) \leq -b$ if $|t| \geq c$.
- (c) The derivative $h'(t)$ exists in some neighbourhood of $t = 0$.
- (d) The second derivative $h''(0)$ exists and $h''(0) < 0$.

Further suppose that the function defined by

$$F(x) = \int_{-\infty}^{\infty} e^{xh(t)} dt$$

converges for all sufficiently large x . Then

$$F(x) \sim \sqrt{2\pi} \left(-xh''(0) \right)^{-1/2}$$

as $x \rightarrow \infty$.

Corollary 1.5.1 (Stirling's formula). *We have*

$$\Gamma(x+1) \sim \sqrt{2\pi} \cdot e^{-x} x^{x+1/2}$$

as $x \rightarrow +\infty$ along the real axis.

Proof. Let $x > 0$. In the integral

$$\Gamma(x+1) = \int_0^\infty e^{-t} t^x dt$$

we replace t with $x(1+t)$ and obtain

$$\Gamma(x+1) = e^x x^{x+1} \int_{-1}^\infty \left(e^{-t} (1+t) \right)^x dt = e^x x^{x+1} \int_{-1}^\infty e^{x(\log(1+t)-t)} dt.$$

Theorem 1.5 with $h(t) = \log(1+t) - t$ completes the proof. $\square$

1.4 Zeta function

Euler was the first to study the function

$$\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s} \tag{1.9}$$

for real values of s strictly greater than 1. Suppose $s > 1$. Then by grouping terms of (1.9) as follows

$$\begin{aligned} \zeta(s) &= \frac{1}{1^s} + \left(\frac{1}{2^s} + \frac{1}{3^s} \right) + \left(\frac{1}{4^s} + \frac{1}{5^s} + \frac{1}{6^s} + \frac{1}{7^s} \right) + \dots \\ &< \frac{1}{1^s} + \frac{2}{2^s} + \frac{4}{4^s} + \dots = \sum_{n=1}^\infty (2^{-(s-1)})^k, \end{aligned}$$

we see that $\zeta(s)$ is finite.

We fix $s = \sigma + it \in \mathbb{C}_{>1} := \{z \in \mathbb{C} : \operatorname{Re}(z) > 1\}$, where σ and t denote the real and imaginary parts of s , respectively. Since $|\sum_{n=1}^\infty n^{-s}| \leq \zeta(\sigma) < \infty$, (1.9) is finite for every $s \in \mathbb{C}_{>1}$.

Analyticity of $\zeta(s)$ on $\mathbb{C}_{>1}$. We denote the N -th partial sum of (1.9) by $\zeta_N(s) := \sum_{n=1}^N \frac{1}{n^s}$. For every positive integer N , $\zeta_N(s)$ is analytic on $\mathbb{C}_{>1}$. In light of Weierstrass's theorem, it suffices to show that ζ_N converges to ζ on every vertical strip² in $\mathbb{C}_{>1}$. Weierstrass's M -test does the job: on the vertical strip given by $1 < \operatorname{Re}(s) < t$, for each n , $|n^{-s}|$ is bounded above by n^{-t} , which are non-negative terms of the convergent series $\zeta(t)$.

Riemann studied the zeta function (and we continue to do so) due to its striking connection to the distribution of primes. The next theorem offers a taste of this connection.

Theorem 1.6 (Euler product). *For $s \in \mathbb{C}_{>1}$, we have that*

$$\zeta(s) = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s} \right)^{-1}. \tag{1.10}$$

²Weierstrass's theorem requires uniform convergence on *every* compact subset of the region in consideration. Any compact subset of $\mathbb{C}_{>1}$ will be bounded, and hence contained in an open vertical strip.

Proof. We first show that the infinite product on the right of (1.10) is analytic. Then it is enough to prove the identity for real $s > 1$.

Analyticity of the infinite product. Since $(1 - p^{-s})^{-1} = 1 + (p^s - 1)^{-1}$, it is enough to show that

$$\sum_{p \text{ prime}}^{\infty} \left| \frac{1}{p^s - 1} \right| \quad (1.11)$$

converges absolutely on compact subsets of $\mathbb{C}_{>1}$.

Any compact subset K of $\mathbb{C}_{>1}$ can be enclosed by a rectangle $R_{a,b,c}$ with vertices at $a + ic$, $a - ic$, $b + ic$ and $b - ic$, where $a, b > 1$ and c are real numbers. By the minimum principle, $|p_s - 1|$ must take its minimum value on the boundary of $R_{a,b,c}$, which is easily shown to be $p^a - 1$. Hence, for $s \in K \subset R_{a,b,c}$, (1.11) is bounded by $\sum_{p \text{ prime}} (p^a - 1)^{-1}$, which converges (by comparison with the convergent series $\sum_{p \text{ prime}} p^{-a}$).

The identity for real $s > 1$. For any prime p , we have the geometric series $(1 - p^{-s})^{-1} = \sum_{k=0}^{\infty} p^{-ks}$. We can then write, by the Fundamental Theorem of Arithmetic,

$$\prod_{p \leq N} (1 - p^{-s})^{-1} = \sum_{n=1}^N n^s + R_N(s),$$

where the product runs over primes p less than or equal to N . Clearly, $R_N(s) \leq \sum_{k=N+1}^{\infty} n^{-s} \rightarrow 0$. $\square$

Corollary 1.6.1. *The zeta function $\zeta(s)$ does not vanish for $s \in \mathbb{C}_{>1}$.*

Proof. Since all the terms of the infinite product in (1.11) do not vanish for each $s \in \mathbb{C}_{>1}$, the infinite product converges to a non-zero number for each $s \in \mathbb{C}_{>1}$. $\square$

We will use the following proposition to study the growth of the zeta function (in §5).

Proposition 1.4. *For $s \in \mathbb{C}_{>1}$,*

$$\zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{x\}}{x^{s+1}} dx \quad (1.12)$$

where $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of x .

Proof. Clearly, for any $s \in \mathbb{C}_{>1}$, the integral in (1.12) is finite as

$$\left| \int_1^{\infty} \frac{\{x\}}{x^{s+1}} dx \right| \leq \int_1^{\infty} x^{-|s+1|} dx < \infty.$$

since the fractional part is less than 1 and $|x^{-(s+1)}| \leq x^{-|s+1|}$. When $x \in [\nu, \nu + 1)$, we have $\{x\} = x - \nu$ so the integral $\int_1^\infty \frac{\{x\}}{x^{s+1}} dx$ is equal to

$$\begin{aligned}
& \sum_{\nu=1}^{\infty} \int_{\nu}^{\nu+1} \frac{x - \nu}{x^{s+1}} dx \\
&= \sum_{\nu=1}^{\infty} \left[\left(\frac{x^{-(s-1)}}{-(s-1)} \right) \Big|_{\nu}^{\nu+1} - \nu \left(\frac{x^{-s}}{-s} \right) \Big|_{\nu}^{\nu+1} \right] \\
&= \sum_{\nu=1}^{\infty} \left[\frac{1}{s-1} \left(\frac{1}{\nu^{s-1}} - \frac{1}{(\nu+1)^{s-1}} \right) - \frac{1}{s} \left(\frac{1}{(\nu+1)^{s-1}} - \frac{1}{\nu^{s-1}} \right) \right] - \frac{1}{s} \sum_{\nu=1}^{\infty} \frac{1}{(\nu+1)^s} \\
&= \frac{1}{s(s-1)} \sum_{\nu=1}^{\infty} \left(\frac{1}{\nu^{s-1}} - \frac{1}{(\nu+1)^{s-1}} \right) - \frac{1}{s} (\zeta(s) - 1) \\
&= \frac{1}{s} \left(\frac{s}{s-1} - \zeta(s) \right),
\end{aligned}$$

which proves (1.12). □

Since the integral in (1.12) defines an analytic function in s for $\operatorname{Re} s > 0$, we actually have (provided that $\zeta(s)$ has been analytically continued to the complex plane) that (1.12) holds for all $s \in \mathbb{C}_+$.

1.5 Jacobi theta function

In 1828, Jacobi [8] proved the following transformation formula.

Theorem 1.7. *For $(z, \alpha) \in \mathbb{C}_+ \times \mathbb{C}$, the following formula holds*

$$z^{1/2} \sum_{n=-\infty}^{\infty} e^{-\pi(n+\alpha)^2 z} = \sum_{n=-\infty}^{\infty} e^{2\pi i n \alpha} e^{-\pi n^2 / z}. \quad (1.13)$$

We will prove this theorem through the Poisson summation formula from Fourier analysis. We only state the essential definitions and result from Chapter 5 of [18], which we recommend for details and proofs.

For $f \in L^1(\mathbb{R})$, the *Fourier transform* of f is given by

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x t} dx.$$

The *Schwartz space* on $\mathbb{R}$, denoted by $\mathcal{S}(\mathbb{R})$, consists of smooth complex-valued functions f on $\mathbb{C}$ with rapidly decreasing derivatives, viz.

$$\sup_{x \in \mathbb{R}} |x|^k |f^{(l)}(x)| < \infty$$

for all $k, l \geq 0$.

Theorem 1.8 (Poisson summation formula). *If $f \in \mathcal{S}(\mathbb{R})$, then*

$$\sum_{n=-\infty}^{\infty} f(x+n) = \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n x}.$$

Proof of Theorem 1.7. We take $f(x) = e^{-\pi x^2 z}$ for $\operatorname{Re} z > 0$. Evidently, $f \in \mathcal{S}(\mathbb{R})$. We compute the Fourier transform of f by completing the square:

$$\begin{aligned} \hat{f}(t) &= \int_{-\infty}^{\infty} e^{-\pi x^2 z} e^{-2\pi i x t} dx \\ &= \int_{-\infty}^{\infty} \exp \left[-\pi \left((x - itz^{-1})^2 + (tz^{-1})^2 \right) z \right] dx \\ &= e^{-\pi t^2 / z} \int_{-\infty}^{\infty} e^{-\pi (x - it/z)^2 z} dx \\ &= e^{-\pi t^2 / z} \int_{-\infty}^{\infty} e^{-\pi x^2 z} dx \\ &= z^{-1/2} e^{-\pi t^2 / z} \int_{-\infty}^{\infty} e^{-\pi x^2} dx \\ &= z^{-1/2} e^{-\pi t^2 / z}. \end{aligned}$$

The theorem follows by Poisson summation. □

We define the (basic) Jacobi theta function as

$$\vartheta(x) := \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x}$$

for $x > 0$. It satisfies the functional equation

$$\vartheta\left(\frac{1}{x}\right) = x^{1/2} \vartheta(x) \tag{1.14}$$

which follows from (1.13) for $\alpha = 0$.

We will work with

$$\theta(x) := \sum_{n=1}^{\infty} e^{-\pi n^2 x} = \frac{\vartheta(x) - 1}{2}$$

and

$$\omega(x) := 3x\theta'(x) + 2x^2\theta''(x), \tag{1.15}$$

both defined for $x > 0$. It follows from (1.14) that

$$\theta\left(\frac{1}{x}\right) = x^{1/2} \theta(x) + \frac{x^{1/2}}{2} - \frac{1}{2}. \tag{1.16}$$

1.6 Functional equation for the zeta function

In this section, we derive the functional equation of the zeta function and use it to extend the zeta function to a meromorphic function on $\mathbb{C}$ with a simple pole at $s = 1$. This approach to the functional equation of the zeta function is one of the two approaches present in Riemann's original paper, and we loosely follow Edwards' translation from [4].

Let $s = \sigma + it \in \mathbb{C}_{>1}$. In the integral

$$\Gamma\left(\frac{s}{2}\right) = \int_0^\infty e^{-x} x^{s/2} \frac{dx}{x},$$

we replace x with $n^2\pi x$ to get

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \frac{1}{n^s} = \int_0^\infty e^{-n^2\pi x} x^{s/2} \frac{dx}{x}.$$

Summing the above equation over all natural numbers n yields

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \sum_{n=1}^\infty \int_0^\infty e^{-n^2\pi x} x^{s/2} \frac{dx}{x} = \int_0^\infty \theta(x) x^{s/2} \frac{dx}{x}. \quad (1.17)$$

The summation and integral could be exchanged because of the absolute convergence of the integrand.

We evaluate the right most side of (1.17) by first splitting the integral over $(0, 1)$ and $[1, \infty)$, replacing x by $1/x$ in the second integral, and then using (1.16):

$$\begin{aligned} \int_0^\infty \theta(x) x^{s/2} \frac{dx}{x} &= \int_1^\infty \theta(x) x^{s/2} \frac{dx}{x} + \int_0^1 \theta(x) x^{s/2} \frac{dx}{x} \\ &= \int_1^\infty \theta(x) x^{s/2} \frac{dx}{x} + \int_1^\infty \theta\left(\frac{1}{x}\right) x^{-s/2} \frac{dx}{x} \\ &= \int_1^\infty \theta(x) x^{s/2} \frac{dx}{x} + \int_1^\infty \left[x^{1/2} \theta(x) + \frac{x^{1/2}}{2} - \frac{1}{2} \right] x^{-s/2} \frac{dx}{x} \\ &= \int_1^\infty \theta(x) [x^{s/2} + x^{(1-s)/2}] \frac{dx}{x} + \frac{1}{2} \int_1^\infty [x^{-(s-1)/2} - x^{-s/2}] \frac{dx}{x}. \end{aligned}$$

Since $\int_1^\infty x^{-a} \frac{dx}{x} = \frac{1}{a}$ for $a > 0$, the second integral is equal to $\frac{1}{s(s-1)}$. Thus, (1.17) along with the above computation proves the identity

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \theta(x) [x^{s/2} + x^{(1-s)/2}] \frac{dx}{x} \quad (1.18)$$

for $\sigma > 1$.

The integral on the right-hand side of (1.18) converges absolutely for all $s \in \mathbb{C}$. Using Morera's theorem, it can be shown that the integral also defines an entire function in s .

We note that

$$\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) \quad (1.19)$$

can be continued to an entire function using (1.18). Because of the symmetry of the right-hand side of (1.18), we have that $\xi(s) = \xi(1-s)$ for all $s \in \mathbb{C}$.

We can now define the analytic continuation of ζ by

$$\zeta(s) = \frac{\xi(s)\pi^{s/2}}{(s-1)\Gamma\left(\frac{s}{2}+1\right)}. \quad (1.20)$$

It is obvious that this new function is analytic on $\mathbb{C} \setminus \{1\}$, with a simple pole at $s = 1$ with residue 1.

We know that $\zeta(s)$ is non-zero for $\sigma > 1$ (Corollary 1.6.1). For $\sigma < 0$, by (1.20), $\zeta(s)$ vanishes precisely for those s for which $\Gamma\left(\frac{s}{2}+1\right)$ has a pole, i.e. when s is a negative even number. These are called the *trivial zeroes* of the zeta function.

Riemann Hypothesis. *The non-trivial zeroes of $\zeta(s)$ have real part $1/2$.*

Moreover, because of (1.19), the number s is a zero of $\xi(s)$ if and only if s is a non-trivial zero of $\zeta(s)$. Hence, the RH can be stated in terms of the Riemann xi function, and we will work with ξ instead of the ζ function.

1.7 Two properties of the Riemann xi function

In this section, we show that the Riemann xi function, which is entire, has order 1 and is real, that is, it is a real-valued function when restricted to the real numbers.

Proposition 1.5. *The entire function ξ has order 1.*

Proof. Since $\xi(s) = \xi(1-s)$, it is enough to establish (1.5) in the half-plane $\sigma \geq 1/2$, where (1.19) holds.

For $\sigma \geq 1/2$, the real logarithm is defined. We have

$$\left| \frac{1}{2}s(s-1) \right| \leq \frac{1}{2}|s|(|s|+1) = e^{\log(1/2)+\log|s|+\log(|s|+1)} \leq e^{c_1|s|}$$

for an some $c_1 > 0$. We used the fact that $\log(1+x) < x$ for $x > 0$. It is also easy to see that $|\pi^{-s/2}| \leq e^{c_2|s|}$ for some $c_2 > 0$ as $|e^s| \leq e^{\operatorname{Re}(s)} \leq e^{|s|}$.

In the integral representation of the zeta function (1.12), the integral is bounded in the half-plane and $|(s-1)^{-1}|$ is bounded for $|s|$ sufficiently large enough. Hence

$$|\zeta(s)| \leq c_3|s|$$

for an appropriate choice of $c_3 > 0$.

Stirling's approximation (Corollary 1.5.1) implies that

$$|\Gamma(s/2)| \leq \Gamma(|s|) \sim \sqrt{2\pi} e^{-|s|} e^{(|s|-1)\log(|s|)}.$$

In particular, there exists a constant $c_4 > 0$ such that

$$|\Gamma(s/2)| \leq e^{c_4|s|\log|s|}.$$

for sufficiently large $|s|$.

Combining all of our estimates, we have that

$$|\xi(s)| \leq e^{c|s|\log|s|}$$

for sufficiently large $|s|$. Since $\log(x) = O(x^\varepsilon)$ as $x \rightarrow \infty$ for each $\varepsilon > 0$, we have proved, for every $\varepsilon > 0$, that $|\xi(s)| = O(e^{|s|^{1+\varepsilon}})$. Since $|\zeta(s)| \rightarrow 1$ as $s \rightarrow \infty$ along the real axis, this does not hold for any negative value of ε . $\square$

Proposition 1.6. *For positive integers n , we have*

$$\xi^{(2n)}(1/2) = 2^{1-2n} \int_1^\infty \omega(x) x^{-3/4} (\log x)^{2n} dx,$$

where

$$\omega(x) = \sum_{n=1}^\infty [2(\pi n^2 x)^2 - 3\pi n^2 x] e^{-\pi n^2 x}$$

as defined in (1.15). In particular, $\xi^{(2n)}(1/2)$ is real for each $n \in \mathbb{N}$.

Proof. (1.9) can be written as

$$\xi(s) = \frac{1}{2} - \frac{s(1-s)}{2} \int_1^\infty \theta(x) (x^{s/2-1} + x^{(1-s)/2-1}) dx.$$

We use integration by parts twice:

$$\begin{aligned} \xi(s) &= \frac{1}{2} + \theta(1) + \int_1^\infty x^{3/2} \theta'(x) ((1-s)x^{(s-1)/2-1} + sx^{-s/2-1}) dx \\ &= \frac{1}{2} + \theta(1) + 4\theta'(1) + \int_1^\infty \frac{d}{dx} [x^{3/2} \theta'(x)] (2x^{(s-1)/2} + 2x^{-s/2}) dx \end{aligned}$$

The functional equation (1.16) proves, on differentiation, that $\frac{1}{2} + \theta(1) + 4\theta'(1) = 0$. Hence,

$$\xi(s) = \int_1^\infty \omega(x) x^{-3/4} \left(x^{\left(\frac{s}{2}-\frac{1}{4}\right)} + x^{-\left(\frac{s}{2}-\frac{1}{4}\right)} \right) dx.$$

The formula is seen to hold on comparison of the coefficients with the power series for $\cosh(x)$. $\square$

Since $\xi(1-s) = \xi(s)$, the odd derivatives of $\xi(s)$ at $s = 1/2$ vanish. Thus, we have the series expansion

$$\xi(s) = \sum_{n=0}^{\infty} \frac{\xi^{(2n)}(1/2)}{(2n)!} (s - 1/2)^{2n}. \quad (1.21)$$

We will work with a slightly modified version:

$$\Theta(z) := \xi(1/2 + \sqrt{z}) = \sum_{m=0}^{\infty} \frac{\gamma(m)}{m!} z^m \quad (1.22)$$

where $\gamma(m) := \frac{m!}{(2m)!} \xi^{(2m)}(1/2)$. It follows from Proposition 1.6 that both (1.21) and (1.22) are *real*, i.e. both map $\mathbb{R}$ to $\mathbb{R}$.

Chapter 2

A Polynomial Miscellany

In the first two sections, we define orthogonal polynomial families with respect to a positive weight function w on an interval I and discuss some results on the location of their zeroes. This general theory is then applied to the case of Hermite polynomials— one of the three classical orthogonal families. Szegő's *Orthogonal Polynomials* [19] is a classic reference, from where the proofs on location of zeroes are taken.

The last two sections offer an exposition of Schur's [17] proof of his composition theorem, which says that the coefficient-wise product of two polynomials with real roots has real roots if the non-zero roots of one of the polynomials have the same sign.

In this chapter, we take the word 'polynomial' to mean polynomial with real coefficients.

2.1 Basic theory and location of zeroes

We say that a sequence of polynomials $\{p_n\}_{n=0}^\infty$ is a *family* if the degree¹ of p_n is n , for each $n \in \mathbb{N}$. If $\{p_n\}$ is a family of polynomials, then any polynomial f of degree m can be written as

$$f = \sum_{k=0}^m \lambda_k p_k. \quad (2.1)$$

where λ_k ($0 \leq k \leq m$) are reals. This is evident if f is either of degree -1 or 0 . We proceed via strong induction on the degree of f . For a polynomial f of degree $m \geq 1$, we choose λ such that $f - \lambda p_m$ has degree at most $m-1$. By the induction hypothesis, $f - \lambda p_m$ is then an $\mathbb{R}$ -linear combination of the polynomials $p_0, p_1, \dots, p_{m-1}$ and the induction goes through.

Let $I = [a, b]$ be an open interval of $\mathbb{R}$. (We will allow a and b to take the values $\pm\infty$.) Let $w : I \rightarrow \mathbb{R}_+$ be an integrable function. A family of polynomials $\{p_n\}_{n=0}^\infty$ is said to be

¹We take the degree of the constant zero polynomial to be -1 , and the non-zero constant polynomials to be 0 .

orthogonal with respect to the weight w if

$$\langle p_m, p_n \rangle_w := \int_a^b p_m(x) p_n(x) w(x) dx = 0 \quad \text{whenever } m \neq n.$$

The function $\langle \cdot, \cdot \rangle_w$ defined above is an inner product. In most cases, the family of orthogonal polynomials will not be normal with respect to this inner product and it will be convenient to keep track of the (squared) norms $\nu_n := \langle p_n, p_n \rangle_w$ and the leading coefficient α_n of $p_n(x)$, for each $n \in \mathbb{N}$. The coefficients λ_n in (2.1) can be determined in terms of the inner product $\langle \cdot, \cdot \rangle_w$. Since

$$\langle f, p_k \rangle = \left\langle \sum_{l=0}^n \lambda_l p_l, p_k \right\rangle = \lambda_k \langle p_k, p_k \rangle,$$

we have $\lambda_k = \nu_k^{-1} \langle f, p_k \rangle$ for $k = 0, 1, \dots, n$.

Example 2.1 (Classical orthogonal polynomials). We mention the weights and intervals with respect to which the following polynomials form an orthogonal family. We only prove, later in §2, that the Hermite polynomials form an orthogonal family. Proofs for the other two families may be found in Rainville [14]. Here α and β are arbitrary but fixed real numbers, and $(x)_k := x(x+1)\dots(x+k-1)$ denotes the rising factorial.

(a) The Jacobi polynomials

$$J_n^{(\alpha, \beta)}(x) = \sum_{k=0}^n \frac{(1+\alpha)_n (1+\beta)_n}{k! (n-k)! (1+\alpha)_k (1+\beta)_{n-k}} \left(\frac{x-1}{2} \right)^k \left(\frac{x+1}{2} \right)^{n-k}$$

with weight $w(x) = (1-x)^\alpha (1+x)^\beta$ on the interval $(-1, 1)$.

(b) The Hermite polynomials

$$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k n! (2x)^{n-2k}}{k! (n-2k)!}$$

with weight $w(x) = e^{-x^2}$ on $\mathbb{R} = (-\infty, \infty)$.

(c) The Legendre polynomials

$$L_n^{(\alpha)}(x) = \sum_{k=0}^n \frac{(-1)^k (1+\alpha)_n x^k}{k! (n-k)! (1+\alpha)_k}$$

with weight $w(x) = x^\alpha e^{-x}$ on the interval $(0, \infty)$.

The roots of an orthogonal family are all real (in fact, lie in the interval I) and simple. Moreover, every orthogonal family satisfies a three-term recurrence relation. These statements are proved below.

Proposition 2.1. *For any polynomial f of degree less than n , $\langle f, p_n \rangle_w = 0$.*

Proof. For $0 \leq m < n$, we write x^m in the form of (2.1) as

$$x^m = \sum_{k=0}^m \lambda_{m,k} p_k(x).$$

We then note that $\langle x^m, p_n \rangle_w = 0$. Since f is an $\mathbb{R}$ -linear combination of these monomials, the proposition follows. $\square$

Proposition 2.2. *For $n > 0$, the polynomial $p_n(x)$ has n distinct roots which all lie in $[a, b]$.*

Proof. Since $\int_a^b p_n(x)w(x)dx = \langle p_n, 1 \rangle = 0$ (Proposition 2.1), $p_n(x)$ has at least one root in (a, b) . Let $x_1, x_2, \dots, x_l$ be the $l \geq 1$ distinct roots of $p_n(x)$ in (a, b) . Define $h(x) := (x - x_1) \dots (x - x_l)$. We note that $p_n(x)h(x)$ is either non-negative or non-positive on (a, b) , and strictly positive or negative on $(a, b) \setminus \{x_1, x_2, \dots, x_l\}$. In either case, we have

$$\langle p_n, h \rangle = \int_a^b p_n(x)h(x)w(x)dx \neq 0.$$

Proposition 2.1 then forces the degree of h to be at least n . But the degree of h can be at most n , and this means that p_n has n distinct roots in (a, b) . $\square$

Proposition 2.3 (Three-term recurrence). *There exist real sequences $\{A_n\}_{n=1}^\infty$, $\{B_n\}_{n=1}^\infty$ and $\{C_n\}_{n=1}^\infty$ such that*

$$xp_n(x) = A_n p_{n+1}(x) + B_n p_n(x) + C_n p_{n-1}(x)$$

for all $n > 0$. If we denote the leading coefficient of p_n by α_n , then

$$A_n = \frac{\alpha_n}{\alpha_{n+1}} \text{ and } C_n = \frac{\nu_n}{\nu_{n-1}} \cdot \frac{\alpha_{n-1}}{\alpha_n}.$$

for all n . In particular, A_n and C_n do not vanish for any n .

Proof. The polynomial $xp_n(x)$ is of degree $n + 1$ and so can be written as the $\mathbb{R}$ -linear combination $\sum_{k=0}^{n+1} \lambda_k p_k(x)$. Since

$$\lambda_k = \nu_k^{-1} \langle xp_n, p_k \rangle_w = \langle p_n, xp_k \rangle_w,$$

we have, by Proposition 2.1, that $\lambda_k = 0$ if $k < n - 1$. Thus, the first part follows by taking $A_n = \lambda_{n+1}$, $B_n = \lambda_n$ and $C_n = \lambda_{n-1}$.

Since

$$\langle x^n, p_n \rangle_w = \int_a^b x^n p_n(x)dx = \alpha_n^{-1} \int_a^b p_n(x)^2 dx = \frac{\nu_n}{\alpha_n}$$

we have that

$$A_n = \lambda_{n+1} = \nu_{n+1}^{-1} \langle xp_n, p_{n+1} \rangle_w = \frac{\alpha_n}{\alpha_{n+1}}$$

and

$$C_n = \lambda_{n-1} = \nu_{n-1}^{-1} \langle xp_n, p_{n+1} \rangle_w = \nu_{n-1}^{-1} \langle p_n, xp_{n+1} \rangle_w = \frac{\nu_n}{\nu_{n-1}} \cdot \frac{\alpha_{n-1}}{\alpha_n}.$$

It is clear that both these quantities do not vanish. $\square$

For a fixed $k > 0$, we can use Proposition 2.3 to write

$$\begin{aligned} (y-x)p_k(x)p_k(y) &= p_k(x)(yp_k(y)) - (xp_k(x))p_k(y) \\ &= A_k(p_{k+1}(y)p_k(x) - p_{k+1}(x)p_k(y)) - C_k(p_k(y)p_{k-1}(x) - p_k(x)p_{k-1}(y)). \end{aligned}$$

Using the relation between A_n and C_n with α_n and ν_n from Proposition 2.3, we have

$$\begin{aligned} \nu_k^{-1}p_k(x)p_k(y) &= \frac{\alpha_k}{\nu_k\alpha_{k+1}} \cdot \frac{p_{k+1}(y)p_k(x) - p_{k+1}(x)p_k(y)}{y-x} \\ &\quad - \frac{\alpha_{k-1}}{\nu_{k-1}\alpha_k} \cdot \frac{p_k(y)p_{k-1}(x) - p_k(x)p_{k-1}(y)}{y-x}. \end{aligned}$$

This identity politely asks to be telescoped and proves:

Proposition 2.4 (Christoffel-Darboux formula).

$$\sum_{k=0}^n \nu_k^{-1}p_k(x)p_k(y) = \frac{\alpha_n}{\nu_n\alpha_{n+1}} \cdot \frac{p_{n+1}(y)p_n(x) - p_{n+1}(x)p_n(y)}{y-x}, \quad (2.2)$$

Since the right-hand side of (2.2) is equal to

$$\frac{\alpha_n}{\nu_n\alpha_{n+1}} \cdot \frac{(p_{n+1}(y) - p_{n+1}(x))p_n(x) - p_{n+1}(x)(p_n(y) - p_n(x))}{y-x},$$

applying limits as $y \rightarrow x$ on both sides of (2.2) yields

$$\sum_{k=0}^n \nu_k^{-1}p_k(x)^2 = \frac{\alpha_n}{\nu_n\alpha_{n+1}} \left(p'_{n+1}(x)p_n(x) - p_{n+1}(x)p'_n(x) \right). \quad (2.3)$$

Suppose $p_n(x)$ and $p_{n+1}(x)$ share a root, say $x = \xi$. Then the right-hand side of (2.3) is zero at $x = \xi$. But the left-hand side of (2.3) is strictly positive for all $x \in I$, and we have a contradiction. This proves the following proposition.

Proposition 2.5. *Two consecutive polynomials of an orthogonal family have no roots in common.*

We have a nice separation theorem for the zeroes of orthogonal polynomials, under the condition that the leading coefficient of each polynomial in the family is positive.

Theorem 2.1. Suppose $\{p_n(x)\}_{n+1}^\infty$ is an orthogonal family of polynomials such that the leading coefficient α_n of each p_n is positive. Let $x_1 < \dots < x_n$ be the roots of $p_n(x)$, then each of the intervals

$$(a, x_1), (x_1, x_2), \dots, (x_{n-1}, x_n), (x_n, b)$$

contains exactly one root of $p_{n+1}(x)$.

Proof. Since the leading coefficients are positive, (2.3) implies that

$$p'_{n+1}(x)p_n(x) - p_{n+1}(x)p'_n(x) > 0. \quad (2.4)$$

Fix $i \in \{1, 2, \dots, n-1\}$. To exploit (2.4), we need to know the sign of $p'_n(x_k)$, which is equal to

$$\begin{aligned} p'_n(x)|_{x=x_k} &= \sum_{k=1}^{\infty} \alpha_n(x-x_1) \dots \widehat{(x-x_k)} \dots (x-x_n) \Big|_{x=x_k} \\ &= (x_k - x_1) \dots (x_k - x_{k-1})(x_k - x_{k+1}) \dots (x_k - x_n) \\ &= (-1)^{n-k} (x_k - x_1) \dots (x_k - x_{k-1})(x_{k+1} - x_k) \dots (x_n - x_k). \end{aligned}$$

So the sign of $p'_n(x_k)$ is $(-1)^{n-k}$. Hence $p'_n(x_k)$ and $p'_n(x_{k+1})$ have opposite signs. The inequality (2.4) for $x = x_k$ (respectively $x = x_{k+1}$) implies that $p_{n+1}(x_k)$ (respectively $p_{n+1}(x_{k+1})$) differs in sign from $p'_n(x_k)$ (respectively $p'_n(x_{k+1})$). Hence, $p_{n+1}(x_k)$ and $p_{n+1}(x_{k+1})$ have opposing signs, and there is at least one root of p_{n+1} in (x_k, x_{k+1}) .

Since the leading coefficient of $p_{n+1}(x)$ is positive, and all roots of $p_{n+1}(x)$ are strictly less than b , we have that $p_{n+1}(b)$ is positive. Since $p'_n(x_n) > 0$, (2.4) implies that $p_{n+1}(x_n) < 0$. Hence, there is at least one zero of p_{n+1} in (x_n, b) .

Similarly, the sign of $p_{n+1}(a)$ is $(-1)^{n+1}$. The sign of $p'_n(x_1)$ is $(-1)^{n-1}$, which implies that the sign of $p_{n+1}(x_1)$ is $(-1)^n$. Hence, there is a root of $p_{n+1}(x)$ in (a, x_1) .

Each of the specified $n+1$ intervals must contain exactly one root as p_{n+1} has $n+1$ distinct roots. $\square$

2.2 Hermite Polynomials

Definition 2.1 (Generating function). We define the n -th Hermite polynomial $H_n(x)$ as the coefficient of $w^n/n!$ in the power series expansion of e^{2xw-w^2} .

Since

$$\begin{aligned}
e^{2xw-w^2} &= e^{2xw} e^{-w^2} \\
&= \left(\sum_{k=0}^{\infty} \frac{(2xw)^k}{k!} \right) \left(\sum_{l=0}^{\infty} \frac{(-1)^l w^{2l}}{l!} \right) \\
&= \sum_{n=0}^{\infty} \left(\sum_{\nu=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\nu (2x)^{n-2\nu}}{\nu! (n-2\nu)!} \right) \frac{w^n}{n!},
\end{aligned}$$

we have that

$$H_n(x) = n! \sum_{\nu=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\nu 2^{n-2\nu}}{\nu! (n-2\nu)!} x^{n-2\nu},$$

which agrees with the expression given in Example 2.1(b). For each $n \in \mathbb{N}$, the degree of $H_n(x)$ is n , and its leading coefficient is 2^n . The first four Hermite polynomials are

$$H_0(x) = 1, \quad H_1(x) = 2x, \quad H_2(x) = 4x^2 - 2, \quad \text{and} \quad H_3(x) = 8x^3 - 12x.$$

We summarize the basic identities for the Hermite polynomials in the following proposition.

Proposition 2.6 (Basic identities). *The following hold for all $n \in \mathbb{N}$.*

(a) *Annihilation identity.*

$$H'_{n+1}(x) = 2nH_n(x),$$

(b) *Creation identity.*

$$H_{n+1}(x) = 2xH_n(x) - H'_n(x), \tag{2.5}$$

(c) *Three-term recurrence.*

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$$

(d) *Christoffel-Darboux formula.*

$$\sum_{k=0}^n \frac{H_k(x)H_k(y)}{2^k k!} = \frac{1}{2^{n+1} n!} \cdot \frac{H_{n+1}(y)H_n(x) - H_{n+1}(x)H_n(y)}{y - x}. \tag{2.6}$$

Proof. Part (a) and (b) follow immediately by comparing coefficients in

$$\sum_{n=0}^{\infty} H'_{n+1}(x) \frac{x^{n+1}}{(n+1)!} = \frac{d}{dx} e^{2xw-w^2} = 2we^{2xw-w^2} \sum_{n=0}^{\infty} (2nH_n(x)) \frac{w^{n+1}}{(n+1)!}$$

and

$$\sum_{n=0}^{\infty} H_{n+1}(x) \frac{w^n}{n!} = \frac{d}{dw} e^{2xw-w^2} = (2x-2w)e^{2xw-w^2} = \sum_{n=0}^{\infty} (2xH_n(x) - H'_n(x)) \frac{w^n}{n!}.$$

respectively. Part (c) follows from part (a) and (b) combined. The Christoffel-Darboux formula can be proved using the three-term recurrence (as done in the last section). $\square$

The following proposition will be used to prove that the Hermite polynomials form an orthogonal family.

Proposition 2.7 (Rodrigues' formula). *The n -th Hermite polynomial is given by the formula*

$$H_n(x) = (-1)^n e^{x^2} D^n e^{-x^2} \quad (2.7)$$

where D denotes the operator $\frac{d}{dx}$.

Proof. We proceed by induction on n . The claim is obvious for $n = 0$ (we interpret D^0 as the identity operator). Suppose (2.7) holds. Then

$$\begin{aligned} DH_n(x) &= (-1)^n D e^{x^2} D^n e^{-x^2} \\ &= (-1)^n (2x e^{x^2} D^n e^{-x^2} + e^{x^2} D^{n+1} e^{-x^2}) \\ &= 2x H_n(x) - (-1)^{n+1} e^{x^2} D^{n+1} e^{-x^2}. \end{aligned}$$

Using (2.5), we get

$$H_{n+1}(x) = (2x - D)H_n(x) = (-1)^{n+1} e^{x^2} D^{n+1} e^{-x^2}.$$

$\square$

Proposition 2.8. *The Hermite polynomials form an orthogonal family with respect to the weight $w(x) = e^{-x^2}$ on $\mathbb{R} = (-\infty, \infty)$. In particular, for non-negative integers m and n ,*

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = \begin{cases} 0 & m \neq n, \\ 2^n n! \sqrt{\pi} & m = n. \end{cases}$$

Proof. If $m \neq n$, then we may assume $m < n$ without loss of generality. (If $m > n$, interchange them.) The Rodrigues' formula implies that $H_n(x) e^{-x^2} = (-1)^n D^n e^{-x^2}$. Hence,

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = (-1)^n \int_{-\infty}^{\infty} H_m(x) D^n e^{-x^2} dx = 0$$

since H_m is a polynomial of degree less than n , repeated partial integration will eventually produce 0. If $m = n$, we write $H_n(x) = 2^n x^n + \pi_{n-1}$, where π_{n-1} denotes an arbitrary polynomial of degree $n - 1$. Then

$$\begin{aligned} \int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx &= (-2)^n \int_{-\infty}^{\infty} x^n D^n e^{-x^2} dx + \int_{-\infty}^{\infty} \pi_{n-1} D^n e^{-x^2} dx \\ &= 2^n n! \int_{-\infty}^{\infty} e^{-x^2} dx = 2^n n! \sqrt{\pi}, \end{aligned}$$

since the integral with the π_{n-1} term vanishes. □

The results of the last section apply immediately, proving the following two theorems.

Theorem 2.2. *For each $k \geq 0$, the roots of $H_k(x)$ are real and simple, i.e. $H_k(x)$ has exactly k distinct real roots.*

Theorem 2.3. *Let $x_1 < \dots < x_k$ be the roots of $H_k(x)$, then each of the intervals*

$$(-\infty, x_1), (x_1, x_2), \dots, (x_{k-1}, x_k), (x_k, \infty)$$

contains exactly one root of $H_{k+1}(x)$. In particular, $H_k(x)$ and $H_{k+1}(x)$ do not have any roots in common.

We end this section with Turán's proof of the following result, which appeared in [20].

Theorem 2.4 (Turán, 1959). *Let $G(x) := \sum_{\nu=0}^d c_\nu H_\nu(x)$ such that each c_ν is real. If*

$$\sum_{k=0}^{d-2} 2^k k! c_k^2 < 2^d (d-1)! c_d^2$$

then the roots of G are real and simple.

Proof. Let $x_1 > x_2 > \dots > x_{n-1}$ denote the distinct roots of $H_{n-1}(x)$. Let x_0 and x_n denote $-\infty$ and ∞ , respectively. Then, for any integer k ($1 \leq k \leq l$), the Christoffel-Darboux formula (2.6) reads

$$\sum_{\nu=0}^{n-2} \frac{H_\nu(x_k) H_\nu(y)}{2^\nu \nu!} = -\frac{1}{2^n (n-1)!} \cdot \frac{H_n(x_k) H_{n-1}(y)}{y - x}.$$

In this equation, if we take the limit as y approaches x_k , the right-hand side will have the term $H'_{n-1}(x_k)$, which, using (2.5), is equal to $-H_n(x_k)$. Tidying up, we have proved

$$\sum_{\nu=0}^{n-2} \frac{H_\nu^2(x_k)}{2^\nu \nu!} = \frac{H_n^2(x_k)}{2^n (n-1)!} \tag{2.8}$$

for $k = 1, 2, \dots, n-1$. Without loss of generality, we assume that c_n , the leading coefficient of G , is positive. If we show² (for $k = 0, 1, \dots, n$) that

$$\text{sign } G(x_k) = (-1)^k, \quad (2.9)$$

we would have shown that G experiences $n+1$ sign changes, and hence has n distinct real roots. Since G is a degree n polynomial, it also shows that these roots are simple. Since $H_n(x) = 2^n x^n + \pi_{n-1}$, (2.9) is obvious for $k = 0$ and $k = n$. By Theorem 2.3, we know that

$$\text{sign } H_n(x_k) = (-1)^k$$

for $k = 0, 1, \dots, n$. And so, for $k = 1, 2, \dots, n-1$, we have $(-1)^k c_n H_n(x_k) = c_n |H_n(x_k)|$. Using Cauchy-Schwarz and (2.8), we get the following inequality

$$\begin{aligned} (-1)^k G(x_k) &= c_n |H_n(x_k)| + \sum_{\nu=0}^{n-2} (-1)^k c_\nu H_\nu(x_k) \\ &\geq c_n |H_n(x_k)| - \sum_{\nu=0}^{n-2} |c_\nu H_\nu(x_k)| \\ &= c_n |H_n(x_k)| - \sum_{\nu=0}^{n-2} \frac{|H_\nu(x_k)|}{2^{\nu/2} \nu!^{1/2}} \cdot 2^{\nu/2} \nu!^{1/2} |c_\nu| \\ &> c_n |H_n(x_k)| - \left(\sum_{\nu=0}^{n-2} \frac{H_\nu(x_k)^2}{2^\nu \nu!} \right)^{1/2} \left(\sum_{\nu=0}^{n-2} 2^\nu \nu! c_\nu^2 \right)^{1/2} \\ &= |H_n(x_k)| \left\{ c_n - \left(\frac{1}{2^n (n-1)!} \sum_{\nu=0}^{n-2} 2^\nu \nu! c_\nu^2 \right)^{1/2} \right\} > 0, \end{aligned}$$

which finishes the proof. □

2.3 Sturm sequences

We outline the bare essentials of *generalized Sturm sequences*, which will be used in the next section to prove Schur's composition theorem. We recommend [13] for a more thorough treatment, from where the following definitions and theorem are taken.

Definition 2.2. Let a_i ($0 \leq i \leq m$) be $m+1$ real numbers. We denote the number of sign changes, as one reads left to right while ignoring zeroes, in $a_0, a_1, \dots, a_m$, by

$$\mathcal{V}(a_0, a_1, \dots, a_m).$$

²By the sign of G at infinity, we mean whether it diverges to positive or negative infinity.

Let f_i ($0 \leq i \leq m$) be $m + 1$ polynomials. For any real x , we use the shorthand

$$\mathcal{V}(x) := \mathcal{V}(f_0(x), f_1(x), \dots, f_m(x)).$$

We also define

$$\mathcal{V}(-\infty) = \lim_{x \rightarrow -\infty} \mathcal{V}(x)$$

and

$$\mathcal{V}(+\infty) = \lim_{x \rightarrow +\infty} \mathcal{V}(x).$$

If a_i is the leading coefficient of f_i (for $1 \leq i \leq m$), then

$$\mathcal{V}(-\infty) = (a_0, -a_1, a_2, \dots, (-1)^m a_m)$$

and

$$\mathcal{V}(+\infty) = (a_0, a_1, a_2, \dots, a_m).$$

Definition 2.3. A sequence of polynomials $f_0, f_1, \dots, f_m$ is a (generalized) Sturm sequence on an interval I if it has the following properties.

- (i) If $\xi \in I$ is a zero of f_0 , then $f_1(\xi) \neq 0$;
- (ii) If $\xi \in I$ is a zero of f_μ and $1 \leq \mu \leq m - 1$, then $f_{\mu-1}(\xi)f_{\mu+1}(\xi) < 0$;
- (iii) $f_m(x) \neq 0$ for all $x \in I$.

We say $\xi \in I$ is a *relevant zero* of f_0 if $f_0(x)$ changes sign at $x = \xi$. We say a relevant zero ξ is of

- the first kind if $f_0(\xi + h)f_1(\xi) > 0$,
- the second kind if $f_0(\xi + h)f_1(\xi) < 0$,

for all sufficiently small positive numbers h .

For a polynomial f , we denote the number of distinct relevant zeros of the first (respectively, second) kind in $[a, b]$ of f by $N_f^+[a, b]$ (respectively, $N_f^-[a, b]$).

Theorem 2.5. Let $f := f_0, f_1, \dots, f_m$ be a Sturm sequence on an interval $[a, b]$. Suppose that f does not vanish at a and b . Then

$$N_f^+[a, b] - N_f^-[a, b] = \mathcal{V}(a) - \mathcal{V}(b).$$

This is proved as Theorem 10.5.3 in [13]. We remark that the theorem still holds if $a = -\infty$ and $b = +\infty$.

2.4 Schur's composition theorem

The assertion that some function has only real zeroes will occur so frequently from this point on that it warrants the following definition.

Definition 2.4. A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is said to be *hyperbolic* if all its zeroes are real. It is said to be *sign-hyperbolic* (or *s-hyperbolic*) if it is hyperbolic and its non-zero zeroes are of the same sign, i.e. either all positive or all negative.

Schur's composition theorem, stated and proved in [17], is an essential component of Pólya and Schur's characterization of the Laguerre-Pólya class of functions which we will discuss in the next chapter.

Theorem 2.6 (Schur's composition theorem, 1914). *Let $f(x) = \sum_{k=0}^m a_k x^k$ and $g(x) = \sum_{k=0}^n b_k x^k$ be two polynomials of degrees m and n respectively, such that f is hyperbolic, and g is s-hyperbolic. Then the polynomial of degree $t = \min(m, n)$, defined by*

$$h(x) = \sum_{k=0}^t k! \cdot a_k b_k x^k,$$

is hyperbolic. Moreover, if $m \leq n$ and $a_0 b_0 \neq 0$, then the roots of h are simple.

Before we proved this theorem, we require three preliminary propositions on hyperbolic polynomials.

Proposition 2.9. *Let $h(x) = \sum_{k=0}^n c_k x^k$ be a hyperbolic polynomial of degree n such that $c_0 \neq 0$. Then $c_{r-1} c_{r+1} < 0$ for $0 < r < n$. In particular, no two consecutive coefficients of $h(x)$ can vanish.*

Proof. We can write

$$h(x) = c_0 \prod_{r=1}^n (1 + \gamma_r x)$$

where c_0 and γ_r ($1 \leq r \leq n$) are non-zero. Since

$$c_1^2 - 2c_0 c_2 = c_0^2 (\gamma_1^2 + \cdots + \gamma_n^2),$$

we have that

$$D_1 := c_1^2 - 2c_0 c_2 > 0.$$

Note that for each r ,

$$h^{(r-1)}(x) = (r-1)! c_{r-1} + \frac{r!}{1!} c_r x + \frac{(r+1)!}{2!} c_{r+1} x^2 + \dots$$

is hyperbolic. If $c_{r-1} \neq 0$, then by the same reasoning used to show $D_1 > 0$, we have that

$$D_r := rc_r^2 - (r+1)c_{r-1}c_{r+1} > 0.$$

If $c_1 = 0$, then $D_1 > 0$ implies that $c_0c_2 < 0$. That is, both c_1 and c_2 cannot be 0. If $c_2 = 0$, then $c_1 \neq 0$, which implies that $D_2 > 0$, and hence $c_1c_3 < 0$. It is clear how this argument may be repeated until we have the desired conclusion. $\square$

Proposition 2.10. *Let $h(x) = \sum_{k=0}^n c_k x^k$ be a hyperbolic polynomial of degree n such that $c_0 \neq 0$. Then*

$$F(x) := h(D)f(x) = \sum_{k=0}^n c_k f^{(k)}(x),$$

where $D = \frac{d}{dx}$ is the differentiation operator, is hyperbolic for any hyperbolic polynomial $f(x)$. Moreover, every multiple root α of F is also a multiple root of f .

Proof. Just like the proof of the previous proposition, we write

$$h(x) = c_0 \prod_{r=1}^n (1 + \gamma_r x)$$

where c_0 and γ_r ($1 \leq r \leq n$) are non-zero. Suppose $f(x)$ is a hyperbolic polynomial of degree m . Of course,

$$F(x) = c_0 \prod_{r=1}^n (1 + \gamma_r D) f(x). \quad (2.10)$$

So it is enough to prove the proposition for $h(x) = 1 + \gamma x$, where γ is a non-zero real. Consider the function $\phi(x) = e^{x/\gamma} f(x)$ which has m real zeroes. By Rolle's theorem,

$$\phi'(x) = \frac{e^{x/\gamma}}{\gamma} (f(x) + \gamma f'(x))$$

has at least $m - 1$ real roots, and thus $h(D)f(x) = f(x) + \gamma f'(x)$ has at least $m - 1$ real roots, and so has exactly m real roots.

Because of (2.10), it is enough to show that if α is a multiple root of $F(x) = (1 + \gamma D)f(x)$ (for $\gamma \neq 0$), then it is a multiple root of $f(x)$. We write the Taylor expansion of $f(x)$ at $x = \alpha$ as

$$f(x) = \sum_{k=0}^m a_k (x - \alpha)^k.$$

Then

$$\begin{aligned} F(x) &= f(x) + \gamma f'(x) \\ &= \sum_{k=0}^m a_k (x - \alpha)^k + \sum_{k=1}^m k\gamma a_k (x - \alpha)^{k-1} \\ &= \sum_{k=0}^m (a_k + (k+1)\gamma a_{k+1}) (x - \alpha)^k. \end{aligned}$$

Since α is a multiple root of F , we must have $a_0 + \gamma a_1 = a_1 + 2\gamma a_2 = 0$, which implies that $a_1^2 - 2a_0a_2 = 0$. Proposition 2.9 then implies $a_0 = 0$, which means that $a_1 = a_2 = 0$. Thus, α is a multiple root of f . $\square$

Proposition 2.11 (Limit of hyperbolic polynomials). *For $n \in \mathbb{N}$, let*

$$p_n(x) = \sum_{k=0}^m a_k^{(n)} x^k$$

define a sequence of hyperbolic polynomials. Suppose, for $k = 0, 1, \dots, m$, that

$$\lim_{n \rightarrow \infty} a_k^{(n)} = a_k.$$

Then $p(x) := \sum_{k=0}^m a_k x^k$ is a hyperbolic polynomial.

Proof. If p happens to be identically zero, then there's nothing to prove. So we may assume that p is a non-zero polynomial. For any compact subset K of the complex plane, we have that $p_n \rightarrow p$ uniformly on K . If we take K to be a compact subset that does not intersect with the real line, then Hurwitz's theorem³ implies that p cannot vanish on K . Hence, the roots of p must lie on the real line in $\mathbb{C}$. $\square$

We are now ready to prove Schur's composition theorem.

Proof of Theorem 2.6. Without loss of generality, we may assume that the leading coefficients of f and g are positive and that the roots of g are negative. This means that the coefficients of g are non-negative.

Suppose, for now, that $a_0b_0 \neq 0$ (i.e. both a_0 and b_0 are non-zero).

The case $m \leq n$. Let s be a fixed non-zero real number. Then we define the polynomial in x ,

$$F(s, x) := g(sD)f(x).$$

Since g is s -hyperbolic, so is the polynomial $x \mapsto g(sx)$, and so by Proposition 2.10, $F(s, x)$ is hyperbolic. If we write

$$F(s, x) = \sum_{k=0}^m \frac{P_k(s)}{k!} x^k,$$

then $P_\mu(s)$ (for $\mu = 0, 1, \dots, m$) must satisfy

$$\frac{P_\mu(s)}{\mu!} = \sum_{k=0}^{m-\mu} b_k(sD)^k a_{\mu+k} x^{\mu+k}.$$

³Suppose $\{f_n\}_{n \in \mathbb{N}}$ is a sequence of analytic functions such that, for each $n \in \mathbb{N}$, f_n does not vanish on Ω . If $f_n \rightarrow f$ on every compact subset of Ω , then if f is not identically equal to 0 on Ω , it does not vanish on any point of Ω . For a proof, see the first section of chapter 5 of [1].

Thus, $P_\mu(s)$ is a polynomial in s given by

$$P_\mu(s) = \sum_{k=0}^{m-\mu} (\mu+k)! a_{\mu+k} b_k s^k.$$

We note that $P_0(s) = h(s)$ and that $P_0(s), P_1(s), \dots, P_m(s)$ is a generalized Sturm sequence (Definition 2.3):

- (i) If $s = \xi$ is a zero of both $P_0(s)$ and $P_1(s)$, then $x = 0$ is a multiple root of F , and by Proposition 2.10, a multiple root of f , which is not possible;
- (ii) If $s = \xi$ is a zero of P_μ and $1 \leq \mu \leq m-1$, then by Proposition 2.9, $P_{\mu-1}(\xi)P_{\mu+1}(\xi) < 0$;
- (iii) The polynomial $P_m(s) = m!a_m b_0$ does not vanish for any s .

Then the number of real roots of $h(s) = P_0(s)$ is at least $\mathcal{V}(-\infty) - \mathcal{V}(+\infty)$.

The leading coefficient of $P_\mu(s)$ is $m!a_m b_{m-\mu}$ which, by our initial assumption, is positive. It is then clear that $\mathcal{V}(-\infty) = m$ and $\mathcal{V}(+\infty) = 0$. So h has at least m real roots. Since h is a polynomial of degree m , it must be hyperbolic. The theorem is thus proved when $a_0 b_0 \neq 0$ and $m \leq n$.

The case $m > n$. We work with the s -hyperbolic polynomials of degree m

$$\left(1 + \frac{x}{\mu}\right)^{m-n} g(x) =: \sum_{k=0}^m b_k^{(\mu)} x^k.$$

By the preceding case, we have that

$$\sum_{k=0}^m k! a_k b_k^{(\mu)} x^k$$

is hyperbolic for each $\mu \in \mathbb{N}^*$. Since

$$\lim_{\mu \rightarrow \infty} b_k^{(\mu)} = \begin{cases} b_k & \text{for } 0 \leq k \leq n, \\ 0 & \text{for } n+1 \leq k \leq m, \end{cases}$$

Proposition 2.11 shows that $\sum_{k=0}^n k! a_k b_k x^k$ is hyperbolic.

Thus, the theorem holds if $a_0 b_0 \neq 0$. What if $a_0 b_0 = 0$? Let r (respectively s) be the smallest integer such that a_r (respectively b_s) is non-zero. We write $f(x) = x^r \tilde{f}(x)$ and $g(x) = x^s \tilde{g}(x)$. For each $\mu \in \mathbb{N}^*$,

$$f_\mu(x) = \left(x + \frac{1}{\mu}\right)^r \tilde{f}(x) = \sum_{k=0}^m a_k^{(\mu)} x^k$$

is a hyperbolic polynomial such that $a_0^\mu \neq 0$ and $\lim_{\mu \rightarrow \infty} a_k^{(\mu)} = a_k$ for $0 \leq k \leq m$. Similarly, for each $\mu \in \mathbb{N}^*$,

$$g_\mu(x) = \left(x + \frac{1}{\mu}\right)^s \tilde{g}(x)$$

is an s-hyperbolic polynomial such that $b_0^{(\mu)} \neq 0$ and $\lim_{\mu \rightarrow \infty} b_k^{(\mu)} = b_k$ for $0 \leq k \leq n$. We set $t = \min(m, n)$. Since, for each $\mu \in \mathbb{N}^*$, $a_0^{(\mu)} b_0^{(\mu)} \neq 0$, we know that

$$\sum_{k=0}^t k! a_k^{(\mu)} b_k^{(\mu)} x^k$$

is hyperbolic. Moreover, $\lim_{\mu \rightarrow \infty} a_k^{(\mu)} b_k^{(\mu)} \rightarrow a_k b_k$ for $0 \leq k \leq t$, and the proof is finished with an application of Proposition 2.11. $\square$

Chapter 3

The Laguerre-Pólya Class

In this chapter, much like the last, we take the word ‘polynomial’ to mean polynomial with real coefficients. We recall from Definition 2.4 that a function is said to be hyperbolic if it only has real zeroes. It is said to be s-hyperbolic if it is hyperbolic and its non-zero zeroes have the same sign.

Definition 3.1. An entire function f is said to be in the *Laguerre-Pólya class* if there exists a sequence of hyperbolic polynomials $\{f_n\}_{n \in \mathbb{N}}$ such that $f_n \rightarrow f$ uniformly on every compact subset of the complex plane. We say that a function f in the Laguerre-Pólya class is of *type I* if there exists a sequence of s-hyperbolic polynomials $\{f_n\}_{n \in \mathbb{N}}$ such that $f_n \rightarrow f$ uniformly on every compact subset of the complex plane.

We denote the class of Laguerre-Pólya functions of type I by $\mathcal{LP}(I)$ and the Laguerre-Pólya class by $\mathcal{LP}$.

The goal of this chapter is to prove Pólya and Schur’s [10] remarkable characterisations¹ of $\mathcal{LP}(I)$ and $\mathcal{LP}$. In the last section, this characterisation is applied to obtain two equivalent formulations of the Riemann Hypothesis.

3.1 Multiplier sequences

Let $\{\gamma_k\}_{k \in \mathbb{N}}$ be a sequence of real numbers. We associate an operator to this sequence, $\Gamma : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$, defined by

$$\Gamma\left(\sum_{k=0}^n a_k x^k\right) := \sum_{k=0}^n \gamma_k a_k x^k,$$

¹The paper by Pólya and Schur refers to $\mathcal{LP}$ as entire functions of type II and $\mathcal{LP}(I)$ as entire functions of type I. This makes sense in light of the fact that entire functions of type I (respectively, II) correspond to multiplier sequences of type I (respectively, II). We prefer the modern terminology (i.e. ‘Laguerre-Pólya class’), but this makes our statement of the characterisation theorems in §3 a little awkward.

which we call the *multiplier operator*.

Definition 3.2 (Multiplier sequences). We say that $\{\gamma_n\}$ is a *multiplier sequence (of type I)* if Γ maps every hyperbolic polynomial to a hyperbolic polynomial.

We say that $\{\gamma_n\}$ is a *multiplier sequence (of type II)* if Γ maps every s-hyperbolic polynomial to a hyperbolic polynomial.

It is obvious that a multiplier sequence of type I is also a multiplier sequence of type II. There exist multiplier sequences of type II which are not of type I (see example 3.2(b)).

Example 3.1 (Multiplier sequences of type I). Each of the following is a multiplier sequence of type I.

- (a) $\gamma_n := n$.
- (b) $\gamma_n := q^{n^2}$ where q is a real such that $|q| \leq 1$.
- (c) $\gamma_n := \frac{1}{(\omega)_n}$ where $(\cdot)_n$ denotes the rising factorial and $\omega > 0$.

Example 3.2 (Multiplier sequences of type II). Each of the following is a multiplier sequence of type II.

- (a) Any sequence given in Example 3.1.
- (b) $\gamma_n := \cos(\lambda + (n-1)\vartheta)$, where λ and ϑ are both real. The sequence γ_n is a multiplier sequence of type I if and only if $\vartheta \in n\mathbb{Z}$.

For an overview of the basic properties of multiplier sequences, please see §1 of [12]. The following proposition, roughly put, says that the termwise limit of a sequence of multiplier sequences is a multiplier sequence.

Proposition 3.1. ² For each $n \in \mathbb{N}$, let $\{\gamma_k^{(n)}\}_{k \in \mathbb{N}}$ is a multiplier sequence of type I (respectively, type II) such that, for each $k \in \mathbb{N}$, $\gamma_k^{(n)}$ converges to γ_k as $n \rightarrow \infty$. Then $\{\gamma_k\}$ is a multiplier sequence of type I (respectively, type II).

Proof. We only prove the theorem for multiplier sequences of type I. The proof for type II sequences is near identical and requires only a minor modification of the argument we now present.

Let $f(x) = \sum_{k=0}^m a_k x^k$ be a hyperbolic polynomial. We wish to show that $(\Gamma f)(x) := \sum_{k=0}^n \gamma_k a_k x^k$ is hyperbolic. If Γf is identically zero, then there's nothing to prove, so we may assume $\gamma_k \neq 0$ for at least one k such that $0 \leq k \leq n$. By hypothesis, we know that

$$f_n(x) := \sum_{k=0}^m \gamma_k^{(n)} a_k x^k$$

²Compare with Proposition 2.11.

is hyperbolic for each $n \in \mathbb{N}$. Since $f_n \rightarrow \Gamma f$ uniformly on every compact subset of the upper half-plane, by Hurwitz's theorem it follows that Γf does not vanish on the upper half-plane. Similarly, Γf does not vanish on the lower half-plane. Hence, Γf is hyperbolic. □

3.2 Jensen polynomials

To the sequence of real numbers $\{\gamma_k\}_{k \in \mathbb{N}}$, we associate the formal power series

$$\Phi(x) := \sum_{k=0}^{\infty} \gamma_k \frac{x^k}{k!}.$$

Definition 3.3 (Jensen polynomials associated to a sequence). We say that the polynomial

$$g_d(\gamma; x) = g_d(\Phi; x) = \sum_{k=0}^d \binom{d}{k} \gamma_k x^k$$

is the d -th *Jensen polynomial* associated to the sequence γ (or the formal power series Φ). Similarly, we say that the polynomial

$$g_d^*(\gamma; x) = g_d^*(\Phi; x) = \sum_{k=0}^d \binom{d}{k} \gamma_k x^{d-k}$$

is the d -th *reciprocal Jensen polynomial* associated to the sequence γ (or the formal power series Φ).

We will drop γ and Φ and write $g_d(x)$ (and $g_d^*(x)$) when the context is clear. The polynomials g_d and g_d^* may not be polynomials of degree d . Nonetheless since $g_d^*(x) = x^d g_d(x^{-1})$, if any one of the two polynomials g_d and g_d^* is hyperbolic, then so is the other one. Similarly, if any one of the polynomials is s-hyperbolic, then so is the other one.

In the following proposition, Φ' denotes the formal derivative of Φ . It is easy to see that Φ' is the generating function of the sequence $\{\gamma_{k+1}\}_{k \in \mathbb{N}}$, i.e. the original sequence shifted to the left by one index.

Proposition 3.2. *For $d > 0$, we have*

$$\frac{d}{dx} g_d(\Phi; x) = d \cdot g_{d-1}(\Phi'; x)$$

and

$$\frac{d}{dx} g_d^*(\Phi; x) = d \cdot g_{d-1}^*(\Phi; x)$$

Proof. We simply note that:

$$\frac{d}{dx} \sum_{k=0}^d \binom{d}{k} \gamma_k x^k = \sum_{k=1}^d \binom{d}{k} \gamma_k (k x^{k-1}) = d \sum_{k=0}^{d-1} \binom{d-1}{k} \gamma_{k+1} x^k$$

and

$$\frac{d}{dx} \sum_{k=0}^d \binom{d}{k} \gamma_k x^{d-k} = \sum_{k=0}^{d-1} \binom{d}{k} \gamma_k (d-k) x^{d-k-1} = d \sum_{k=0}^{d-1} \binom{d-1}{k} \gamma_k x^{(d-1)-k}.$$

□

The following proposition is due to O'Sullivan [9].

Proposition 3.3. *Let $\Omega \in \mathbb{R}[[x]]$. For all $d \geq 0$,*

$$g_d^*(\Phi \cdot \Omega; x) = \sum_{k=0}^d \binom{d}{k} \gamma_k g_{d-k}^*(\Omega; x).$$

Proof. Suppose Ω is associated the sequence $\{\delta_n\}_{n \in \mathbb{N}}$. We note that $\Phi \cdot \Omega$ is associated to the sequence $\{\sum_{k=0}^n \gamma_k \delta_{n-k}\}_{n \in \mathbb{N}}$. Using the fact that

$$\binom{d}{n} \binom{n}{k} = \binom{d}{k} \binom{d-k}{n-k},$$

we write

$$\begin{aligned} g_d^*(\Phi \cdot \Omega; x) &= \sum_{n=0}^d \binom{d}{n} \left(\sum_{k=0}^n \binom{n}{k} \gamma_k \delta_{n-k} \right) x^{d-n} \\ &= \sum_{k=0}^d \binom{d}{k} \gamma_k \sum_{n=k}^d \binom{d-k}{n-k} \delta_{n-k} x^{d-n} \\ &= \sum_{k=0}^d \binom{d}{k} \gamma_k \sum_{m=0}^{d-k} \binom{d-k}{m} \delta_m x^{(d-k)-m} \\ &= \sum_{k=0}^d \binom{d}{k} \gamma_k g_{d-k}^*(\Omega; x). \end{aligned}$$

□

3.3 Pólya and Schur's characterisation theorems

We use the convention established in the last section: associated to the sequence of real numbers $\{\gamma_n\}_{n \in \mathbb{N}}$ is the formal power series

$$\Phi(x) = \sum_{k=0}^{\infty} \gamma_k \frac{x^k}{k!}$$

and the d -th Jensen polynomial

$$g_d(\Phi; x) = \sum_{k=0}^d \binom{d}{k} \gamma_k x^k.$$

Theorem 3.1 (Pólya-Schur). *The following are equivalent.*

- (i) $\Phi \in \mathcal{LP}(I)$.
- (ii) There exist some $\rho > 0$ and a sequence of s -hyperbolic real polynomials $\{f_n\}_{n \in \mathbb{N}}$ which converges to Φ uniformly on $\overline{B}(0, \rho)$.
- (iii) $\{\gamma_k\}$ is a multiplier sequence of type I.
- (iv) For each $n \in \mathbb{N}$, g_n is s -hyperbolic.
- (v) The formal power series Φ is an s -hyperbolic real entire function of the form

$$\Phi(x) = \alpha x^m e^{\beta x} \prod_{n=1}^{\infty} (1 + \delta_n x)$$

where $m \in \mathbb{N}$, $\alpha \in \mathbb{R}$ and $\beta \geq 0$.

Proof. (i) $\Rightarrow$ (ii). Obvious.

(ii) $\Rightarrow$ (iii). Suppose $G_n(z) = \sum_{k=0}^n \gamma_k^{(n)} z^k$ is a sequence of s -hyperbolic polynomials which converges to $\Phi(z)$ uniformly on $\overline{B}(0, \rho)$. Hence, $\gamma_k^{(n)} \rightarrow \gamma_k$ as $n \rightarrow \infty$ for each $k \in \mathbb{N}$. By the composition theorem (Theorem 2.6), it follows that for each $n \in \mathbb{N}$, $\{\gamma_k^{(n)}\}_{k \in \mathbb{N}}$ is a multiplier sequence of type I. By Proposition 3.1, it follows that $\{\gamma_k\}$ is a multiplier sequence of type I.

(iii) $\Rightarrow$ (iv). Obvious.

(iv) $\Rightarrow$ (v). Without loss of generality, we may assume that (i) the roots of g_n are non-positive for each $n \in \mathbb{N}$ and (ii) Φ is not identically equal to zero (since the zero function is in $\mathcal{LP}(I)$). Let r be the smallest integer such that $\gamma_r \neq 0$. We write

$$g_n(x) = \binom{n}{r} \gamma_r x^r \prod_{k=1}^{n-r} (1 + \delta_{nk} x)$$

where δ_{nk} is non-negative for $1 \leq k \leq n - r$. On comparing coefficients, we have

$$\sum_{k=1}^{n-r} \delta_{nk} = \frac{\binom{n}{r+1} \gamma_{r+1}}{\binom{n}{r} \gamma_r} = \frac{n-r}{r+1} \cdot \frac{\gamma_{r+1}}{\gamma_r} \quad (3.1)$$

and

$$\prod_{k=1}^{n-r} \delta_{nk} = \frac{\gamma_n}{\binom{n}{r} \gamma_r}.$$

Applying AM-GM yields

$$\frac{\gamma_n}{n!} \leq \frac{1}{(n-r)!} \cdot \frac{\gamma_r}{r!} \left(\frac{\gamma_{r+1}}{(r+1)\gamma_r} \right)^{n-r},$$

since r is fixed, $\Phi(x)$ converges absolutely (this follows, for instance, by comparing $\Phi(x)$ with the series for the exponential function) and thus is an entire function.

The sequence of s-hyperbolic polynomials defined by

$$g_n\left(\frac{x}{n}\right) = \sum_{\nu=0}^n \binom{n}{\nu} \gamma_\nu \frac{x^\nu}{\nu!} = \sum_{\nu=0}^n \left(1 - \frac{1}{m}\right) \left(1 - \frac{2}{m}\right) \cdots \left(1 - \frac{\nu-1}{n}\right) \gamma_\nu \frac{x^\nu}{\nu!}$$

converges uniformly to $\Phi(x)$ on any bounded (and so also compact) set. We take C to be a bounded set which does not intersect with $(-\infty, 0]$. Since the polynomials $g_n(x/n)$ do not vanish on C and converge uniformly to $\Phi(x)$ on C as $n \rightarrow \infty$, Hurwitz's theorem implies that $\Phi(x)$ does not vanish on C either. Hence, the only roots of $\Phi(x)$, if they exist, are non-positive, i.e. Φ is s-hyperbolic.

Suppose $\Phi(x)$ has infinitely many positive roots, which we denote by

$$\frac{1}{\delta_1}, \frac{1}{\delta_2}, \dots$$

The equation (3.1) can be rewritten as the sum

$$\sum_{\nu=1}^n \frac{\delta_{n\nu}}{n} = \left(1 - \frac{r}{n}\right) \cdot \frac{\gamma_{r+1}}{(r+1)\gamma_r},$$

which we use to show that the series

$$\sum_{\nu=1}^{\infty} \delta_\nu \leq \lim_{n \rightarrow \infty} \sum_{\nu=1}^n \frac{\delta_{n\nu}}{n} = \frac{\gamma_{r+1}}{(r+1)\gamma_r}$$

is finite. We are in a position to apply Theorem 1.1. since the order of Φ is ≤ 1 , we have that the genus, which is equal to $\max(k, \deg p)$, is either equal to 0 or 1. Since $k = 0$ (as shown by (3.3)), the theorem follows.

(v) $\Rightarrow$ (i). Suppose

$$G(x) = \alpha x^r e^{\beta x} \prod_{k=1}^{\infty} (1 + \delta_k x)$$

where $\beta \geq 0$ and all non-zero δ_k have the same sign. Then the s-hyperbolic polynomials

$$G_n(x) = \alpha x^r \left(1 + \frac{\beta x}{n}\right)^n \prod_{k=1}^n (1 + \delta_k z)$$

converge uniformly to G on every bounded (and so every compact) subset of the complex plane. $\square$

We also have the following characterisation theorem for the Laguerre-Pólya class, whose proof is not too different.

Theorem 3.2 (Laguerre-Pólya). *The following are equivalent.*

- (i) $\Phi \in \mathcal{LP}$.
- (ii) *There exist some $\rho > 0$ and a sequence of hyperbolic real polynomials $\{f_n\}_{n \in \mathbb{N}}$ which converges to Φ uniformly on $\overline{B}(0, \rho)$.*
- (iii) $\{\gamma_k\}$ *is a multiplier sequence of type II.*
- (iv) *For each $d \in \mathbb{N}$, g_d is hyperbolic.*
- (v) *The formal power series Φ is a hyperbolic entire function of the form*

$$\Phi(x) = \alpha x^m e^{-\beta_1 x^2 + \beta_2 x} \prod_{k=1}^{\infty} (1 + \delta_k x) e^{\delta_k x}$$

where $m \in \mathbb{N}$, $\beta_1 \geq 0$ and $\beta_2 \in \mathbb{R}$.

Proof. We only prove (iv) $\Rightarrow$ (v). The other parts are proved very similarly to the characterisation theorem for $\mathcal{LP}$.

If Φ is not identically equal to zero, there's nothing to prove. So let r be the smallest integer such that $\gamma_r \neq 0$. We write, just as before,

$$g_n(x) = \binom{n}{r} \gamma_r x^r \prod_{k=1}^{n-r} (1 + \delta_{nk} x).$$

On comparing coefficients of x^{r+1} and x^{r+2} we obtain

$$\binom{n}{r} \gamma_r \sum_{\nu=1}^n \delta_{n\nu} = \binom{n}{r+1} \gamma_{r+1}$$

and

$$\binom{n}{r} \gamma_r \sum_{\nu=1, \nu' \neq 1}^{n-r} \delta_{n\nu} \delta_{n\nu'} = \binom{n}{r+2} \gamma_{r+2}$$

respectively. But then

$$\binom{n}{r}^2 \gamma_r^2 \sum_{\nu=1}^{n-r} \delta_{n\nu}^2 = \binom{n}{r+1}^2 \gamma_{r+1}^2 - 2 \binom{n}{r} \binom{n}{r+2} \gamma_{r+2}^2.$$

and

$$\binom{n}{r} \gamma_r \prod_{\nu=1}^{n-r} \delta_{n\nu} = \gamma_n.$$

Applying AM-GM, we get

$$\left(\prod_{\nu=1}^{n-r} \delta_{n\nu}\right)^{1/(n-r)} \leq \frac{1}{n-r} \sum_{\nu=1}^{n-r} \delta_{n\nu}^2,$$

which is the same as

$$\gamma_n^2 \leq \binom{n}{r}^2 \gamma_r^2 \left\{ \frac{\binom{n}{r+1}^2 \gamma_{r+1}^2 - 2 \binom{n}{r} \binom{n}{r+2} \gamma_r \gamma_{r+2}}{(n-r) \binom{n}{r}^2 \gamma_r^2} \right\}^{n-r}.$$

Thus, there exists a constant c (that only depends on r , γ_r , γ_{r+1} and γ_{r+2}) such that

$$\gamma_n^2 \leq n^n c^n.$$

This implies that

$$\frac{|\gamma_n|}{n!} \leq \frac{n^{n/2}}{n!} c^{n/2}. \quad (3.2)$$

This shows that Φ is absolutely convergent (and hence entire).

The order of Φ is at most 2. Because of (3.2), the order of Φ can at most be the order of the entire function

$$g(z) = \sum_{n=0}^{\infty} \frac{n^{n/2}}{n!} z^n.$$

We recall a result³ from Hadamard theory: the order of an entire function $f(z) = \sum_{n=0}^{\infty} a_n z^n$ is related to its coefficients by

$$\lambda = \limsup_{n \rightarrow \infty} \frac{n \log n}{\log(1/|a_n|)}.$$

The order of g can then easily be seen to be at most 2. And so the genus of Φ is at most 2.

We note that the s-hyperbolic polynomials (in x^2) $g_n(x/n)g_n(-x/n)$ converge uniformly to $\Phi(x)\Phi(-x)$ on compact sets. Then Theorem 3.2 implies that $\Phi(x)\Phi(-x)$ is in $\mathcal{LP}(I)$. Thus, Φ is hyperbolic. Moreover, the required product formula follows from the fact that $\sum_{\nu}^{\infty} \gamma_{\nu}^2$ converges (since $1/\gamma_{\nu}^2$ are the zeroes of $\Phi(x)\Phi(-x)$). $\square$

3.4 Applications to the Riemann Hypothesis

After Jensen's death in 1925, Pólya had come across his collected manuscripts in September 1926. In 1927, he published [11] which contains a possible line of attack on the Riemann Hypothesis through the study of the zeroes of Jensen polynomials associated to the Riemann xi function.

³Theorem 2.2.2 from Boas [2].

We recall that

$$\Theta(z) := \xi(1/2 + \sqrt{z}) = \sum_{m=0}^{\infty} \frac{\gamma(m)}{m!} z^m$$

where $\gamma(m) = \frac{m!}{(2m)!}$ for $m \in \mathbb{N}$, defines an entire function of order $1/2$. Proposition 1.6 implies that Θ is a real function. We note that the Riemann Hypothesis holds if and only if the only zeroes of Θ are negative. Thus, the RH holds if and only if any one of the equivalent conditions of Theorem 3.1 hold with $G = \Theta$.

We define the d -th Jensen polynomial of shift n for $\{\gamma(m)\}_{m \in \mathbb{N}}$ as

$$J^{d,n}(x) = \sum_{k=0}^d \binom{d}{k} \gamma(n+k) x^k.$$

Of course, for $n = 0$, $J^{d,n}$ is the d -th Jensen polynomial defined earlier.

Theorem 3.3. *The Riemann Hypothesis is true if and only if $J^{d,n}$ is hyperbolic for all $d \geq 1, n \geq 0$.*

Proof. Suppose the RH holds, then by our earlier discussion (which is essentially an appeal to Theorem 3.1) $J^{d,0}$ is hyperbolic for each $d \geq 0$. Then using Proposition 3.2 we write

$$J^{d,n}(\Theta; x) = g_d(\Theta^{(n)}; x) = \frac{1}{(d+n)(d+n-1) \dots (d+1)} \frac{d^n}{dx^n} g_{d+n}(\Theta; x)$$

which shows that $J^{d,n}$ is hyperbolic as the derivative of the hyperbolic polynomial $g_{d+n}(\Phi; x) = J^{d+n,0}(x)$ is hyperbolic.

The other direction follows immediately from Theorem 3.1, as $J^{d,0}$ is the d -th Jensen polynomial of Θ . $\square$

Similar to the Griffin, Ono, Rolin and Zagier's paper [6], O'Sullivan [9] studies the modified Jensen polynomials

$$P^{d,n}(x) = \sum_{k=0}^d \binom{d}{k} \gamma(n+k) H_{d-k}(x).$$

where $H_d(x)$ denotes the d -th Hermite polynomial, for which we have the following theorem.

Theorem 3.4. *The Riemann hypothesis holds if and only if $P^{d,n}(x)$ is hyperbolic for all $d \geq 1, n \geq 0$.*

We require a proposition from [10] to prove this theorem, which in a sense is the extension of Proposition 2.10 to 'infinite' polynomials.

Theorem 3.5 (Pólya, 1915). *Let $\Phi(z) = \sum_{k=0}^{\infty} \frac{\alpha_k}{k!} z^k$ and $\Psi(z) = \sum_{k=0}^{\infty} \frac{\beta_k}{k!} z^k$ be two functions in $\mathcal{LP}$. Then*

$$\Omega(z) = \sum_{k=0}^{\infty} \frac{\alpha_k}{k!} \Psi^{(k)}(z)$$

converges absolutely and represents a function in $\mathcal{LP}$.

Proof. If we define, for $k \in \mathbb{N}$,

$$\omega_k = \sum_{l=0}^{\infty} \frac{\alpha_l \beta_{l+k}}{l!}$$

then

$$\Omega(z) = \sum_{k=0}^{\infty} \frac{\omega_k}{k!} z^k.$$

Thus, because of Theorem 3.2, it is enough to show that $\{\omega_k\}_{k \in \mathbb{N}}$ is a multiplier sequence of type II.

Consider the polynomials

$$\Phi_n(z) := \sum_{k=0}^n \binom{n}{k} \alpha_k \left(\frac{z}{n}\right)^k$$

and

$$\Psi_n(z) := \sum_{k=0}^n \binom{n}{k} \beta_k \left(\frac{z}{n}\right)^k,$$

which are both hyperbolic for each $n \in \mathbb{N}$ as they are the Jensen polynomials of Φ and Ψ , respectively, evaluated at z/n instead of z . Proposition 2.10 implies that

$$\Phi_n(D)\Psi_n(z) = \sum_{k=0}^n \binom{n}{k} \frac{\Psi_n^{(k)}(z)}{n^k} =: \sum_{k=0}^{\infty} \frac{\omega_k^{(n)}}{k!} z^k$$

is hyperbolic for each $n \in \mathbb{N}$. Hence, $\{\omega_k^{(n)}\}_{k \in \mathbb{N}}$ is a multiplier sequence of type II for each $n \in \mathbb{N}$.

Since $\omega_k^{(n)} \rightarrow \omega_k$ as $n \rightarrow \infty$, an appeal to Proposition 3.1 finishes the proof. $\square$

Proof. As we have already seen, the RH holds if and only if the Riemann xi function is in the Laguerre-Pólya class, which is the case if and only if $\Theta^{(n)}(z)e^{-z^2/2}$ is in the Laguerre-Pólya class for each $n \in \mathbb{N}$. This happens if and only if the reciprocal Jensen polynomials

associated to $\Theta^{(n)}(z)e^{-z^2/2}$ are hyperbolic for all $n \geq 0, d \geq 0$. Since

$$\begin{aligned} g_d^*(e^{-z^2/2}; x) &= \sum_{k=0}^{\lfloor d/2 \rfloor} \binom{d}{2k} (-1)^k \frac{(2k)!}{k!} x^{d-k} \\ &= \sum_{k=0}^{\lfloor d/2 \rfloor} \frac{k!(-1)^k}{(d-2k)!k!} x^{d-k} = H_d(x/2), \end{aligned}$$

by Proposition 3.3, we have

$$g_d^*(\Theta^{(n)}(z)e^{-z^2/2}; x) = \sum_{k=0}^d \binom{d}{k} \gamma(k+n) H_d(x/2) = P^{d,n}(x/2).$$

□

In the next chapter, we use asymptotics for the Riemann xi function to show, for any fixed $d \geq 1$ and n sufficiently large, $P^{d,n}(x)$ is hyperbolic.

Chapter 4

Asymptotics for the Riemann xi function

The objective of this chapter is to prove Griffin, Ono, Rolin and Zagier's [6] asymptotic expansion for $\xi^{(2n)}(1/2)$, which was essential to their proof of Theorem 5.1.

We closely follow O'Sullivan's [9] paper. We first prove a general asymptotic expansion for sequences of the form

$$I_\alpha(f; n) = \int_1^\infty (\log t)^n e^{-\alpha t} f(t) dt \quad (4.1)$$

where $\alpha > 0$ and f satisfies suitable conditions (that are described in §3). This general expansion is then applied to the coefficients of the Riemann xi function to obtain the expansion of Griffin, Ono, Rolin and Zagier.

4.1 The integral form of $\xi^{(2n)}(1/2)$ revisited

In this section, we express $\xi^{(2n)}(1/2)$ as a linear combination of terms of the form (4.1). This will allow us to apply O'Sullivan's expansion to the coefficients of the Riemann xi function.

We recall from Proposition 1.6 that

$$\xi^{(2n)}(1/2) = 2^{1-2n} \int_1^\infty (\log x)^{2n} x^{-3/4} \omega(x) dx \quad (4.2)$$

where

$$\omega(x) = 3x\theta'(x) + 2x^2\theta''(x)$$

and

$$\theta(x) = \sum_{n=1}^{\infty} e^{-\pi n^2 x}$$

both defined for $x > 0$. It follows from definition that

$$\begin{aligned}\omega(x) &= -3\pi x \sum_{n=1}^{\infty} n^2 e^{-\pi n^2 x} + 2\pi^2 x^2 \sum_{n=1}^{\infty} n^4 e^{-\pi n^2 x} \\ &= -3\pi x e^{-\pi x} + 2\pi^2 x^2 e^{-\pi x} + O(x^2 e^{-4\pi x}).\end{aligned}$$

This combined with (4.2) lets us write

$$\xi^{(2n)}(1/2) = 2^{1-2n} (2\pi^2 I_{\pi, 5/4}(2n) - 3\pi I_{\pi, 1/4}(2n) + O(I_{4\pi, 5/4}(2n))) \quad (4.3)$$

where

$$I_{\alpha, \beta}(n) := \int_1^{\infty} (\log t)^n e^{-\alpha t} t^{\beta} dt \quad (4.4)$$

for each $n \in \mathbb{N}$. This latter function is of the form (4.1) with the choice $f(t) := t^{\beta}$.

4.2 Preliminaries for O'Sullivan's expansion

In this section, we collect and prove preliminary lemmas and bounds essential to the proof of O'Sullivan's main result from [9].

4.2.1 Lambert W function

We consider the function $w : [0, \infty) \rightarrow [0, \infty)$ defined by

$$w(x) = x e^x.$$

We note that the derivative of $w(x)$ is strictly positive

$$w'(x) = (1+x)e^x > 0$$

for $x \in [0, \infty)$. It follows that w is an injective map. Since $w(0) = 0$ and $\lim_{x \rightarrow \infty} w(x) = \infty$, w takes every value in $[0, \infty)$. Thus, w is a bijective map.

The inverse of w exists, which we denote by W . This function is called the *Lambert W function*. The following properties of the Lambert W function on $[0, \infty)$ readily follow.

- (i) For $x \in [0, \infty)$, we have $W(xe^x) = x$ and $W(x)e^{W(x)} = x$.
- (ii) $W(0) = 0$ and W is positive on $(0, \infty)$. The latter property follows from the fact that $w|_{[0, \infty)}$ is a bijection.

(iii) $W(x)$ is differentiable¹ on $[0, \infty)$ and increasing on $(0, \infty)$. Since the derivative of w on $(0, \infty)$ is positive,

$$W'(w(x)) = \frac{1}{w'(x)} > 0.$$

for every $x \in [0, \infty)$. Since w is surjective, it follows that the derivative of W is positive on $(0, \infty)$. Hence, W is an increasing function on $(0, \infty)$. This, combined with property (ii) implies that W is increasing on $[0, \infty)$.

(iv) For $x \geq e$, we have $W(x) \leq \log(x)$ since

$$W(x) \leq W(x \log(x)) = W(\log(x)e^{\log(x)}) = \log(x).$$

We plot the graphs $y = xe^x = w(x)$ and $y = W(x)$.

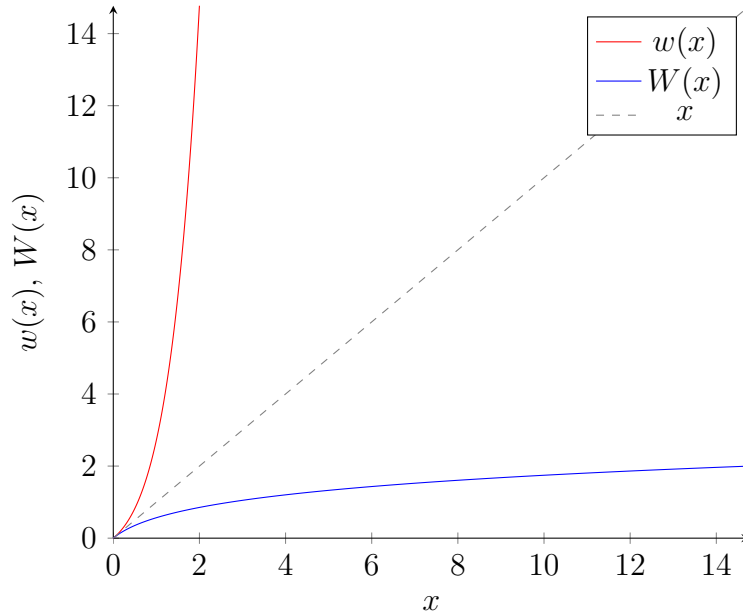


Figure 4.1: The graph of $y = w(x)$ and its inverse function $y = W(x)$.

Proposition 4.1. For $x > 0$ and $c \geq 1$,

$$\left(1 - \frac{1}{W(x)}\right) \log c \leq W(cx) - W(x) \leq \log c.$$

Proof. Since $W(x)e^{W(x)} = x$ and $W(cx)e^{W(cx)} = cx$, taking logarithms on both sides of these equations gives us

$$\log W(x) + W(x) = \log x$$

¹Actually, W is differentiable at 0 as well, but we would have to start with a slightly larger domain of definition for w .

and

$$\log W(cx) + W(cx) = \log c + \log x.$$

Subtracting the first equation from the second, and rearranging, we get

$$W(cx) - W(x) - \log c = \log \frac{W(x)}{W(cx)}.$$

Since W is increasing, the right-hand side of the above equation is ≤ 0 . This proves the right inequality.

We note that

$$\log \frac{W(cx)}{W(x)} \leq \log \left(1 + \frac{\log c}{W(x)} \right) \leq \frac{\log c}{W(x)}$$

since $\log(1+x) \leq x$ for $x \geq 0$. Hence

$$-\frac{\log c}{W(x)} \leq -\log \frac{W(cx)}{W(x)} = \log \frac{W(x)}{W(cx)} = W(cx) - W(x) - \log c,$$

which proves the left inequality. $\square$

4.2.2 The function $h_u(x)$

For $u \geq 2$, we define

$$h_u(x) = \log \left(1 + \frac{\log(1+x)}{u} \right). \quad (4.5)$$

Using the inequality

$$|\log(1+z)| \leq -\log(1-|z|),$$

which is valid for $|z| < 1$, we see that (4.5) defines holomorphic function of x in $B(0, r)$, where $r = 1 - e^{-2} > 3/4$. The *partial ordinary Bell polynomial* $\widehat{\mathcal{B}}_{i,j}$, defined by the generating function

$$(p_1x + p_2x^2 + p_3x^3 + \dots)^j = \sum_{i=j}^{\infty} \widehat{\mathcal{B}}_{i,j}(p_1, p_2, p_3, \dots)x^i,$$

allows us to write the power series expansion of $h_u(x)$ about $x = 0$ using the Taylor series of $\log(1+z)$:

Proposition 4.2 (Taylor series). *For $u \geq 2$ and $|x| \leq 3/4$, we have that*

$$h_u(x) = \sum_{i=1}^{\infty} \ell_i(u)x^i$$

where

$$\ell_i(u) = \sum_{j=1}^i \widehat{\mathcal{B}}_{i,j}(1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots) \frac{(-1)^{j+1}}{j \cdot u^j}. \quad (4.6)$$

Proof. We have already seen that $h_u(x)$ is holomorphic on $B(0, 1 - e^{-2})$. The power series expansion follows from

$$\begin{aligned}
h_u(x) &= \sum_{j=1}^{\infty} \frac{(-1)^{j+1}}{j} \left(\frac{\log(1+x)}{u} \right)^j \\
&= \sum_{j=1}^{\infty} \frac{(-1)^{j+1}}{j u^j} \left(1 - \frac{x}{2} + \frac{x^2}{3} - \dots \right)^j \\
&= \sum_{j=1}^{\infty} \frac{(-1)^{j+1}}{j u^j} \sum_{i=j}^{\infty} \widehat{\mathcal{B}}_{i,j}(1, -\tfrac{1}{2}, \tfrac{1}{3}, \dots) x^i \\
&= \sum_{i=1}^{\infty} \ell_i(u) x^i,
\end{aligned}$$

□

It is easy to compute

$$\ell_1(u) = \frac{1}{u}, \quad \ell_2(u) = -\frac{1}{2u} - \frac{1}{2u^2}. \quad (4.7)$$

Proposition 4.3 (Taylor remainder). *Suppose $u \geq 2$ and $|x| < 1/2$. Then for each $k \geq 0$,*

$$h_u(x) = \sum_{j=1}^k \ell_j(u) x^j + R_k(u, x)$$

where

$$|\ell_j(u)| \leq \frac{3}{u} \left(\frac{4}{3} \right)^j \quad \text{and} \quad |R_k(u, x)| \leq \frac{12}{u} \left(\frac{4}{3} \right)^k |x|^{k+1}.$$

Proof. The inequality $|\log(1+w)| \leq 2|w|$ (for $|w| \leq 3/4$, $w \in \mathbb{C}$) implies

$$|h_u(x)| = \left| \log \left(1 + \frac{\log(1+x)}{u} \right) \right| \leq \frac{2|\log(1+x)|}{u} \leq \frac{4|x|}{u},$$

which for $|x| = 3/4$, gives us the estimate

$$|h_u(x)| \leq \frac{3}{u} \quad (4.8)$$

for $|x| \leq 3/4$. For each $j \in \mathbb{N}$, Cauchy's integral formula implies that

$$h_u^{(j)}(0) = \frac{j!}{2\pi i} \oint_C \frac{h_u(x)}{x^{j+1}} dx$$

where C is the circle of radius $3/4$ centered at the origin (oriented anticlockwise). Since $\ell_j(u) = \frac{h_u^{(j)}(0)}{j!}$ is the coefficient of x^j in the power series of $h_u(x)$ about $x = 0$, it follows that

$$\begin{aligned}
|\ell_j(u)| &= \left| \frac{h_u^{(j)}(0)}{j!} \right| \leq \frac{1}{2\pi} \oint_C \frac{|h_u(x)|}{|x|^{j+1}} dx \\
&\leq \frac{3}{u} \left(\frac{4}{3} \right)^{j+1} \frac{1}{2\pi} \oint_C |dx|
\end{aligned}$$

where we used the bound (4.8) for the second inequality. Since the above integral evaluates to $2\pi \cdot \frac{3}{4}$, we have established one of the two promised bounds:

$$\ell_j(u) \leq \frac{3}{u} \left(\frac{4}{3}\right)^j.$$

We now prove the bound for the remainder term. Since, for $|x| \leq 1/2$,

$$\frac{4}{3}|x| \leq \frac{2}{3}$$

and

$$\frac{1}{1 - \frac{4}{3}|x|} \leq 3,$$

we have that

$$\begin{aligned} |R_k(u, x)| &= \sum_{i=k+1}^{\infty} |\ell_i(u) x^i| \\ &\leq \frac{3}{u} \left(\frac{4}{3}\right)^{k+1} |x|^{k+1} \sum_{i=0}^{\infty} \left(\frac{4}{3}|x|\right)^i \\ &= \frac{3}{u} \left(\frac{4}{3}\right)^{k+1} |x|^{k+1} \cdot \frac{1}{1 - \frac{4}{3}|x|} \\ &\leq \frac{12}{u} \left(\frac{4}{3}\right)^k |x|^{k+1}. \end{aligned}$$

□

Proposition 4.4 (Inequalities for $h_u(x)$). *Let $u \geq 2$. As a function of a real variable x , $h_u(x)$ satisfies the following inequalities:*

(i) for $x \geq 3u$,

$$h_u(x) \leq \frac{x}{2u},$$

and

(ii) for $0 < \delta \leq \frac{1}{100}$, we have that

$$h_u(x) - \frac{x}{u} \leq -\frac{\delta^2}{4u}$$

for all real $x \in (-\infty, -\delta) \cup (\delta, \infty)$.

Proof. The first part of the proposition follows from the inequalities:

$$\log(1+x) \leq x \quad (x \geq 0)$$

and

$$\log(1+x) < \frac{x}{2} \quad (x \geq 3).$$

Then if $x \geq 3u$, we have

$$h_u(x) \leq \log \left(1 + \frac{x}{u} \right) < \frac{x}{2u}.$$

Now we prove the second part. First note that $h_u(x) - x/u$ is increasing for $x < 0$ and decreasing for $x > 0$. This may be verified by checking the first derivative.

Thus, for $x \geq \delta$, we have, using (4.7) and the remainder estimate from Proposition 4.3,

$$\begin{aligned} h_u(x) - \frac{x}{u} &\leq h_u(\delta) - \frac{\delta}{u} \\ &= \frac{\delta}{u} - \left(\frac{1}{2u} + \frac{1}{2u^2} \right) \delta^2 + R_2(u, \delta) - \frac{\delta}{u} \\ &= -\frac{u+1}{2u^2} \delta^2 + |R_2(u, \delta)| \\ &= -\frac{\delta^2}{2u} (1 - 50\delta) \leq -\frac{\delta^2}{4u} \end{aligned}$$

since $\delta \leq 1/100$. Using a similar argument, we obtain the same bound for $x \leq -\delta$. $\square$

4.2.3 Bounds for the incomplete gamma function

The incomplete gamma function is defined as

$$\Gamma(s, a) := \int_a^\infty e^{-x} x^{s-1} dx$$

for positive reals s and a . The convergence of $\Gamma(s, a)$ follows from the convergence of $\Gamma(s)$.

Proposition 4.5. *For $a, r > 0$, we have*

$$\Gamma(r+1, a) \leq 2^r (a^r + \Gamma(r+1)) e^{-a}.$$

Proof. We replace the variable x with $a+x$ and break the integral on the intervals $(0, a)$ and $[a, \infty)$:

$$\begin{aligned} \Gamma(r+1, a) &= \int_a^\infty e^{-x} x^r dx \\ &= \left(\int_0^a e^{-x} (a+x)^r dx + \int_a^\infty e^{-x} (a+x)^r dx \right) e^{-a} \\ &\leq \left(\int_0^a e^{-x} (2a)^r dx + \int_a^\infty e^{-x} (2x)^r dx \right) e^{-a} \\ &\leq \left((2a)^r \int_0^\infty e^{-x} dx + 2^r \int_0^\infty e^{-x} x^r dx \right) e^{-a} \\ &= 2^r (a^r + \Gamma(r+1)) e^{-a}. \end{aligned}$$

$\square$

Proposition 4.6. For all $a, r \geq 0$ and $c > 0$

$$\int_a^\infty e^{-cx} x^r dx \ll c^{-r-1} ((ac)^r + 1) e^{-ac}$$

where the implied constant depends only on r .

Proof. We replace the variable x with cx in the following integral and use the previous proposition:

$$\begin{aligned} \int_a^\infty e^{-cx} x^r dx &= c^{-r-1} \int_{ac}^\infty e^{-x} x^r dx = c^{-r-1} \Gamma(r+1, ac) \\ &\leq c^{-r-1} 2^r ((ac)^r + \Gamma(r+1)) e^{-ac} \\ &\ll c^{-r-1} ((ac)^r + 1) e^{-ac}. \end{aligned}$$

□

4.3 Suitable functions

We focus our attention on the following class of functions which have a well-behaved power series expansion about a point whose terms do not grow too fast.

Definition 4.1. Following O’Sullivan [9], we say that a continuous function $f : [1, \infty) \rightarrow \mathbb{R}_+$ is *suitable* if there exists constant $b, \lambda \geq 0$ and functions $\{f_m\}_{m \in \mathbb{N}}$ which satisfy the following properties:

(i) $f(t) \ll t^b$,

(ii) for $v \geq 1$,

$$f(e^v(1+x)) = f(e^v) \sum_{m=0}^{k-1} f_m(v) x^m + O(v^{\lambda k} |x|^k)$$

in $|x| \leq 1/(2v^\lambda)$ and

(iii) for each $m \in \mathbb{N}$, $f_m(v) \ll v^{\lambda m}$.

The following proposition shows that functions of the form $f(t) := t^\beta$ are suitable (for any $\beta \in \mathbb{R}$). Thus, we see that O’Sullivan’s result (which we state and prove in the next section) applies to (4.4).

Proposition 4.7. The function $f(t) = t^\beta$ is suitable for any $\beta \in \mathbb{R}$ with choice of constants $b = \beta, \lambda = 0$ and functions $f_m(v) = \binom{\beta}{m}$.

Proof. Newton’s binomial theorem implies that

$$f(e^v(1+x)) = (e^v(1+x))^\beta = e^{\beta v} (1+x)^\beta = f(e^v) \sum_{m=0}^{\infty} \binom{\beta}{m} x^m$$

for $|x| < 1$.

□

4.4 O'Sullivan's expansion

In this section, we prove O'Sullivan's asymptotic expansion (Theorem 4.1) for

$$I_\alpha(f; n) = \int_1^\infty (\log t)^n e^{-\alpha t} f(t) dt \quad (4.9)$$

where $n \in \mathbb{N}$, $\alpha > 0$ and f is a suitable function (for the constants b , λ and functions f_m). This is the same sequence as (4.1) which was introduced at the beginning of this chapter.

Theorem 4.1 (O'Sullivan). *Set $u := W(n/\alpha)$. Suppose that f is a suitable function (for the constants b , λ and the functions f_m ($m \in \mathbb{N}$)). Then as $n \rightarrow \infty$ we have*

$$I_\alpha(f; n) = \sqrt{2\pi} \frac{u^{n+1} f(e^u) e^{u-n/u}}{\sqrt{(1+u)n}} \left(1 + \sum_{r=1}^{R-1} \frac{a_r(f; u)}{n^r} + O\left(\frac{u^{R(1+2\lambda)}}{n^R}\right) \right) \quad (4.10)$$

where, for each $r > 0$,

$$a_r(f; v) = \sum_{j=0}^{2r} \frac{(2j+2r-1)!!}{j!} \left(\frac{v^2}{v+1} \right)^{j+r} \sum_{i=j}^{2r} \widehat{\mathcal{B}}_{i,j}(\ell_3(v), \ell_4(v), \dots) \cdot f_{2r-i}(v)$$

is a rational function in v . Here $\widehat{\mathcal{B}}_{i,j}$ are the Bell polynomials defined in §4.2.2 and ℓ_i is as given in (4.6).

The implied constant in (4.10) depends only on $R \geq 1$, α and b . We also have $a_r(f; u) \ll u^{r(1+2\lambda)}$.

Proof. We define, for each $n \in \mathbb{N}$, $g_n(t) := (\log t)^n e^{-\alpha t}$ for $t \in (0, \infty)$. We note that

$$g'_n(t) = \left(\frac{n}{t \log t} - \alpha \right) g_n(t)$$

for $t > 0$. Since $g_n(t)$ is positive on $(1, \infty)$, $g'_n(t) = 0$ only when $t = u$. Since $g_n(t) \rightarrow 0$ as $t \rightarrow \infty$, $t = u$ is the unique point where $g_n(t)$ attains its maximum on $(1, \infty)$.

Step 0. The setup. We will use Laplace's method (see Theorem 1.5 and the discussion preceding it) and expand the integral $I_\alpha(f; n)$ about the point $t_0 = \frac{n}{\alpha u}$.

Since

$$\begin{aligned} \log \left(\frac{g_n(t_0(1+x))}{g_n(t_0)} \right) &= \log \left(\frac{(\log t_0(1+x))^n e^{-\alpha t_0(1+x)}}{(\log t_0)^n e^{-\alpha t_0}} \right) \\ &= \log \left(\left(\frac{\log t_0 + \log(1+x)}{\log t_0} \right)^n \cdot e^{-\alpha t_0 x} \right) \\ &= n \left[h_u(x) - \frac{x}{u} \right], \end{aligned}$$

we have that

$$\begin{aligned}
I_\alpha(f; n) &= \int_1^\infty g_n(t) f(t) dt \\
&= g_n(t_0) \int_1^\infty \exp\left(\log\left(\frac{g_n(t)}{g_n(t_0)}\right)\right) f(t) dt \\
&= t_0 g_n(t_0) \int_{1/t_0-1}^\infty \exp\left(n\left[h_u(x) - \frac{x}{u}\right]\right) f(t_0(x+1)) dx,
\end{aligned}$$

where we made the change of variables $t = t_0(1+x)$.

Step 1. Neglecting the tails. We prove that for $0 < \delta \leq \frac{1}{100}$,

$$\begin{aligned}
&\int_{1/t_0-1}^\infty \exp\left(n\left[h_u(x) - \frac{x}{u}\right]\right) f(t_0(x+1)) dx \\
&= \int_{-\delta}^\delta \exp\left(n\left[h_u(x) - \frac{x}{u}\right]\right) f(t_0(x+1)) dx + O\left(u \cdot n^b \exp\left(-\frac{\delta^2 n}{4u}\right)\right)
\end{aligned}$$

where the implied constant only depends on α and b .

We will consider the integral $I_\alpha(f; n)$ on the following four intervals:

$$I_\alpha(f; n) = t_0 g(t_0) \left(\int_{1/t_0-1}^{-\delta} + \int_{-\delta}^\delta + \int_\delta^{3u} + \int_{3u}^\infty \right) \exp\left(n\left[h_u(x) - \frac{x}{u}\right]\right) f(t_0(x+1)) dx.$$

Proposition 4.4(i) combined with the fact that $f(t) \ll t^b$ shows that

$$\int_{3u}^\infty \left(n\left[h_u(x) - \frac{x}{u}\right]\right) f(t_0(x+1)) dx \ll u \cdot n^{b-1} \exp\left(-\frac{3}{2}n\right),$$

where the implied constant depends only on α and b . Proposition 4.4(ii) combined with the fact that $f(t) \ll t^b$ shows that

$$\left(\int_{1/t_0-1}^{-\delta} + \int_\delta^{3u}\right) \left(n\left[h_u(x) - \frac{x}{u}\right]\right) f(t_0(x+1)) dx \ll u \cdot n^b \exp\left(-\frac{\delta^2 n}{4u}\right),$$

where the implied constant again depends only on α and b . This last bound is larger than the previous bound, and this step is complete.

Step 2. We prove that

$$\begin{aligned}
&\int_{-\delta}^\delta \exp\left(n\left[h_u(x) - \frac{x}{u}\right]\right) f(t_0(x+1)) dx \\
&= \int_{-\delta}^\delta e^{-Cx^2/2} \exp\left(n \sum_{i=3}^n \ell_i(u) x^i\right) f(t_0(x+1)) dx + O(n^{3/5} - n^{2k/5})
\end{aligned}$$

where $C = \frac{n(1+u)}{u^2}$ and $\delta = n^{-2/5}$. We use the Taylor expansion of $h_u(x)$ (Proposition 4.2) and the fact that $n\ell_2(u) = -C/2$:

$$\begin{aligned} & \exp\left(n\left[h_u(x) - \frac{x}{u}\right]\right) \\ &= \exp\left(n\ell_2(u)x^2 + n\sum_{i=3}^k \ell_i(u)x^i + nR_k(u, x)\right) \\ &= e^{-Cx^2/2} \exp\left(n\sum_{i=3}^k \ell_i(u)x^i\right) \exp(O(n\delta^{k+1})). \end{aligned}$$

Since $\delta = n^{-2/5}$ it follows that $\exp(O(n\delta^{k+1})) = 1 + O(n\delta^{k+1})$ and

$$\int_{-\delta}^{\delta} e^{-Cx^2/2} \exp\left(n\sum_{i=3}^k \ell_i(u)x^i\right) f(t_0(x+1)) dx \ll_k 1.$$

This completes this step.

Step 3. We prove, for $|x| \leq \delta := n^{-2/5}$,

$$\exp\left(\sum_{i=3}^k n \cdot \ell_i(u)x^i\right) \frac{f(t_0(1+x))}{f(t_0)} = \sum_{r=0}^{k-1} \varepsilon_r(n, u)x^r + O\left(\frac{1}{n^{k/15}}\right)$$

where the implied constant depends only on k and λ and

$$\varepsilon_r(n, u) := \sum_{j=0}^r e_j(n, u) f_{r-j}(u). \quad (4.11)$$

In this expression, the f_{r-j} 's are the functions associated to the suitable function f and $e_r(n, u)$ is the coefficient of x^r in the expansion of

$$\exp\left(\sum_{i=3}^{\infty} n \cdot \ell_i(u)x^i\right).$$

More explicitly,

$$e_r(n, u) = \sum_{j=0}^{\lfloor r/3 \rfloor} \widehat{\mathcal{B}}_{r-2j,j}(\ell_3(u), \ell_4(u), \dots) \frac{n^j}{j!}.$$

We collect the bounds required to complete this step. It is straightforward, from the definition of Bell polynomials, to prove that

$$e_r(n, u) \ll_r n^{r/3}.$$

We also have the expansion

$$\exp\left(\sum_{i=3}^k n \cdot \ell_i(u)x^i\right) = \sum_{r=0}^{k-1} e_r(n, u)x^r + O\left(\frac{1}{n^{k/15}}\right). \quad (4.12)$$

where $|x| \leq \delta$ and $u \geq 1$. This is Lemma 4.4 of [9].

Since $u = W(n/\alpha) = \log n - \log(\alpha u)$, we have that $u \geq \frac{1}{\alpha}$ implies $u \leq \log n$. Thus, for large enough n , $n^{-2/5} \leq 1/(2u^\lambda)$. So we can substitute $t = t_0$ and $v = u$ in the definition of a suitable function (Definition 4.1) to get

$$\frac{f(t_0(x+1))}{f(t_0)} = \sum_{m=0}^{k-1} f_m(u)x^m + O\left(\frac{u^{\lambda k}}{n^{2k/5}}\right). \quad (4.13)$$

From the bound for $\ell_i(u)$ from Proposition 4.2, it follows that

$$\exp\left(\sum_{i=3}^k n \cdot \ell_i(u)x^i\right) = O(1) \quad (4.14)$$

where the implied constant depends only on k and from (4.13) (with $k = 1$) it follows that

$$\frac{f(t_0(x+1))}{f(t_0)} = O(1). \quad (4.15)$$

The expansions (4.12) and (4.13) along with the bounds (4.14) and (4.15) implies

$$\begin{aligned} \exp\left(\sum_{i=3}^k n \cdot \ell_i(u)x^i\right) \frac{f(t_0(x+1))}{f(t_0)} &= \left[\sum_{r=0}^{k-1} e_r(n, u)x^r + O\left(\frac{1}{n^{k/15}}\right)\right] \left[\sum_{m=0}^{k-1} f_m(u)x^m + O\left(\frac{u^{\lambda k}}{n^{2k/5}}\right)\right] \\ &= \left(\sum_{r=0}^{k-1} e_r(n, u)x^r\right) \left(\sum_{m=0}^{k-1} f_m(u)x^m\right) + O\left(\frac{1}{n^{k/15}} + \frac{u^{\lambda k}}{n^{2k/5}}\right) \\ &= \left(\sum_{r=0}^{k-1} e_r(n, u)x^r\right) \left(\sum_{m=0}^{k-1} f_m(u)x^m\right) + O\left(\frac{1}{n^{k/15}}\right). \end{aligned}$$

It then follows, from (4.11), that

$$\begin{aligned} &\exp\left(\sum_{i=3}^k n \cdot \ell_i(u)x^i\right) \frac{f(t_0(x+1))}{f(t_0)} \\ &= \sum_{r=0}^{k-1} \varepsilon_r(n, u)x^r + \sum_{r=k}^{2k-2} x^r \sum_{j=r-k+1}^{k-1} e_j(nu)f_{r-j}(u) + O\left(\frac{1}{n^{k/15}}\right). \end{aligned} \quad (4.16)$$

Finally, the bound

$$\begin{aligned} \sum_{r=k}^{2k-2} x^r \sum_{j=r-k+1}^{k-1} e_j(nu)f_{r-j}(u) &\ll \sum_{r=k}^{2k-2} \delta^r \sum_{j=r-k+1}^{k-1} n^{j/3} u^{\lambda(r-j)} \\ &\ll \sum_{r=k}^{2k-2} \delta^r n^{(k-1)/3} u^{\lambda(k-1)} \\ &\ll \delta^k n^{(k-1)/3} u^{\lambda(k-1)} \\ &= n^{-k/15-1/3} u^{\lambda(k-1)} \\ &\ll n^{-k/15} \end{aligned}$$

combined with (4.16) completes this step.

Step 4. Central approximation. We prove that

$$\begin{aligned} \int_{-\delta}^{\delta} e^{-Cx^2/2} \exp \left(n \sum_{i=3}^k \ell_i(u) x^i \right) \frac{f(t_0(x+1))}{f(t_0)} dx \\ = \frac{\sqrt{2\pi}u}{\sqrt{(u+1)n}} \sum_{m=0}^{\lfloor (k-1)/2 \rfloor} \frac{a_m^*(f)}{n^m} + O \left(\frac{1}{n^{k/15}} \right), \end{aligned}$$

where

$$a_m^*(f) = \sum_{j=0}^{\lfloor 2m/3 \rfloor} c_{m,j}(f) n^j$$

and

$$c_{m,j}(f) = \left(\frac{(2m-1)!!}{j!} \left(\frac{u^2}{u+1} \right)^m \right) \sum_{i=j}^{2m-2j} \widehat{\mathcal{B}}_{i,j}(\ell_3(u), \ell_4(u), \dots) f_{2m-2j-i}(u).$$

First, we use the last step to obtain

$$\begin{aligned} \int_{-\delta}^{\delta} e^{-Cx^2/2} \exp \left(\sum_{i=3}^k n \cdot \ell_i(u) x^i \right) \frac{f(t_0(1+x))}{f(t_0)} dx \\ = \sum_{r=0}^{k-1} \varepsilon_r(n, u) \int_{-\delta}^{\delta} e^{-Cx^2/2} x^r dx + O \left(\frac{1}{n^{k/15}} \right). \end{aligned}$$

The integral above is zero if r is odd. When r is even, it differs from

$$\int_{-\infty}^{\infty} e^{-Cx^2/2} x^{2m} dx = \sqrt{\frac{2\pi}{C}} \frac{(2m-1)!!}{C^m}$$

by

$$2 \int_{\delta}^{\infty} e^{-Cx^2/2} x^r dx \leq 2 \int_{\delta}^{\infty} e^{-C\delta x/2} x^r dx \ll e^{-n^{1/5}/(2u)}$$

where we used the fact that $C > n/u$ and Proposition 4.6. Thus,

$$\begin{aligned} \int_{-\delta}^{\delta} e^{-Cx^2/2} \exp \left(n \sum_{i=3}^k \ell_i(u) x^i \right) dx \\ = \sqrt{\frac{2\pi}{C}} \sum_{m=0}^{\lfloor (k-1)/2 \rfloor} \varepsilon_{2m}(n, u) \frac{(2m-1)!!}{C^m} + O \left(\frac{1}{n^{k/15}} + n^{k/3} e^{-n^{1/5}/(2u)} \right) \\ = \frac{\sqrt{2\pi}u}{\sqrt{(u+1)n}} \sum_{m=0}^{\lfloor (k-1)/2 \rfloor} \frac{a_m^*(f)}{n^m} + O \left(\frac{1}{n^{k/15}} \right). \end{aligned}$$

The $n^{k/3}$ term comes in the second line from the fact that $\varepsilon_r(n, u) \ll_r n^{r/3}$.

Final step. We replace k with $2Lk + 1$. Then we can write

$$\sum_{m=0}^{\lfloor (k-1)/2 \rfloor} \frac{a_m^*(f)}{n^m} = \sum_{m=0}^{Lk} \frac{a_m^*(f)}{n^m} = \sum_{r=0}^{Lk} \frac{\widehat{a}_r(u)}{n^r},$$

where

$$\widehat{a}_r(u) = \sum_{j=0}^{2r} c_{r+j,j}(f)$$

with the understanding that $c_{m,j}$ is zero for $m > Lk$. Step 4 can then be written as

$$\begin{aligned} \int_{-\delta}^{\delta} e^{-Cx^2/2} \exp \left(n \sum_{i=3}^{2Lk+1} \ell_i(u) x^i \right) \frac{f(t_0(x+1))}{f(t_0)} dx \\ = \frac{\sqrt{2\pi}u}{\sqrt{(u+1)n}} \sum_{r=0}^{Lk} \frac{\widehat{a}_r(f)}{n^r} + O \left(\frac{1}{n^{2Lk/15}} \right). \end{aligned}$$

This combined with Step 2 produces

$$\begin{aligned} \frac{1}{f(t_0)} \int_{-\delta}^{\delta} \exp \left(n \left[h_u(x) - \frac{x}{u} \right] \right) dx \\ = \frac{\sqrt{2\pi}u}{\sqrt{(u+1)n}} \left[\sum_{r=0}^{k-1} \frac{\widehat{a}_r(f)}{n^r} + \sum_{r=k}^{Lk} \frac{\widehat{a}_r(f)}{n^r} + O \left(\frac{n^{1/2}}{n^{(4Lk-1)/5}} + \frac{n^{1/2}}{n^{2Lk/15}} \right) \right]. \end{aligned}$$

The coefficient $\widehat{a}_r(f)$ equals $a_r(f)$ when $r < l$ and $L \geq 3$. If we choose $L = 15$, then

$$\int_{-\delta}^{\delta} \exp \left(n \left[h_u(x) - \frac{x}{u} \right] \right) f(t_0(x+1)) dx = \frac{\sqrt{2\pi}u}{\sqrt{(u+1)n}} f(t_0) \left[\sum_{r=0}^{k-1} \frac{a_r(f)}{n^r} + O \left(\frac{u^k}{n^k} \right) \right]$$

Since $t_0 = e^u$ and $g(t_0) = u^n e^{-n/u}$, the theorem follows. $\square$

4.5 Asymptotic expansion of $\xi^{(2n)}(1/2)$

We end this chapter with a proof of the asymptotic expansion for the coefficients of the powers series (about $1/2$) of the Riemann xi function.

Theorem 4.2. *Set $w := W(2n/\pi)$. As $n \rightarrow \infty$,*

$$\xi^{(2n)}(1/2) = 4\pi^2 e^{7w/2} \sqrt{\frac{2w}{w+1}} \left(\frac{w}{2e^{1/w}} \right)^{2n} \left(1 + \sum_{k=1}^{K-1} \frac{\mu_k(w)}{n^k} + O \left(\frac{\log^K(n)}{n^K} \right) \right)$$

where the implied constant only depends on K . Each $\mu_k(w)$ is a rational function of w with size $O(\log^k(n))$.

Proof. We apply Theorem 4.1 with $f_\beta(t) = t^\beta$, which is a suitable function for any $\beta \in \mathbb{R}$ (Proposition 4.7), to get that

$$I_{\alpha,\beta}(n) = \sqrt{2\pi} \frac{u^{n+1} e^{u(\beta+1)-n/u}}{\sqrt{(1+u)n}} \left(1 + \sum_{r=1}^{R-1} \frac{a_r(f; u)_\beta}{n^r} + O\left(\frac{u^R}{n^R}\right) \right)$$

where $I_{\alpha,\beta}(n)$ is defined as in (4.4) and $u := W(n/\alpha)$. Since

$$\frac{\pi w}{2n} e^w = 1,$$

we have that

$$\begin{aligned} & 2\pi^2 I_{\pi, 5/4}(2n) - 3\pi I_{\pi, 1/4}(2n) \\ &= 2\pi^{5/2} \frac{w^{2n+1} e^{9w/4-2n/w}}{\sqrt{(w+1)n}} \left(1 + \sum_{r=1}^{R-1} \frac{\mu_r(w)}{n^r} + O\left(\frac{w^R}{n^R}\right) \right) \end{aligned} \quad (4.17)$$

where

$$\mu_r(w) := 2^{-r-1} (2a_r(f; w)_{5/4} - 3w \cdot a_{r-1}(f, w)_{1/4}) \ll w^r.$$

Since $e^{w/2} = \sqrt{2n/(\pi w)}$, (4.17) can be rearranged to the form described in the theorem. □

Chapter 5

Hyperbolicity of modified Jensen polynomials

We recall that

$$\Theta(z) := \xi(1/2 + \sqrt{z}) = \sum_{m=0}^{\infty} \gamma(m) \frac{z^m}{m!}$$

is an entire function, which is hyperbolic if and only if the Riemann hypothesis is true. Its m -th coefficient is $\gamma(m) = \frac{m!}{(2m)!} \xi^{(2m)}(1/2)$.

We have proved that the Riemann hypothesis is equivalent to any one of the following statements.

- The Jensen polynomial

$$J^{d,n}(x) = \sum_{k=0}^d \binom{d}{k} \gamma(n+k) x^k$$

is hyperbolic for each $n, d \geq 0$.

- The modified Jensen polynomial

$$P^{d,n}(x) = \sum_{k=0}^d \binom{n}{k} \gamma(n+k) H_{d-k}(x)$$

is hyperbolic for each $n, d \geq 0$. Here H_k (for $k \in \mathbb{N}$) denotes the k -th Hermite polynomial (Definition 2.1).

The first equivalence is given by Theorem 3.3; the second by Theorem 3.4.

These two equivalences naturally suggest a line of attack on the Riemann hypothesis. One hopes to reach ‘closer’ to the Riemann hypothesis by showing that for ‘most’ choices of the integers d and n , the polynomials $J^{d,n}(x)$ and $P^{d,n}(x)$ are hyperbolic.

In this chapter, we present two recent results of this kind. The first result is on the hyperbolicity of $J^{d,n}(x)$; the second on the hyperbolicity of $P^{d,n}(x)$. We also present O’Sullivan’s [9] proof of the latter result, which uses the asymptotic expansion obtained in Theorem 4.2 and Turán’s lemma (Theorem 2.4).

5.1 Jensen polynomials

In 2019, using asymptotics for the Riemann xi function, Griffin, Ono, Rolin and Zagier [6] proved that for any $d \geq 1$, $J^{d,n}(x)$ is hyperbolic for all sufficiently large n .

In Theorem 1.1 of a follow-up paper from 2022, Griffin, Ono, Rolin, Thorner, Tripp and Wagner [7] made this result precise:

Theorem 5.1 (Griffin et al., 2022). *There exists a constant $c > 0$ such that $J^{d,n}(x)$ is hyperbolic whenever $n \geq ce^d$.*

We show below that it is sufficient to consider $P^{d,n}(x)$ instead.

Proposition 5.1 (O’Sullivan). *If for some choice of non-negative integers d and n , we have that $J^{d,n}(x)$ is hyperbolic, then so is $P^{d,n}(x)$.*

Proof. Since $J^{d,n}(x) = \sum_{k=0}^d \binom{d}{k} \gamma(k+n) x^k$ is hyperbolic, so is the reciprocal polynomial

$$J_*^{d,n}(x) := x^d J^{d,n}(x^{-1}) = \sum_{k=0}^d \binom{d}{k} \gamma(k+n) x^{d-k}.$$

Since e^{-x^2} is in the Laguerre-Pólya class, Theorem 3.5 implies that

$$\Omega(x) := \sum_{k=0}^d \binom{d}{k} \gamma(k+n) \frac{d^k}{dx^k} e^{-x^2}$$

is hyperbolic. Rodrigues’ formula (2.7) and the fact that $H_k(x)$ is even when k is even and odd when k is odd implies that $P^{d,n}(x) = \Omega(-x)$ is hyperbolic. $\square$

5.2 O’Sullivan’s modified Jensen polynomials

Turán’s lemma, which was proved in chapter 2, gives us a criterion for the hyperbolicity of $P^{d,n}(x)$ in terms of the coefficients of the power series (about $1/2$) of the Riemann xi function.

Proposition 5.2. *If*

$$\sum_{j=2}^d \frac{d}{2^j j!} \binom{d}{j} \frac{\gamma(n+j)^2}{\gamma(n)^2} < 1,$$

then $P^{d,n}(x)$ is hyperbolic.

Proof. We may write

$$P^{d,n}(x) = \sum_{\nu=0}^d c_\nu H_\nu(x) \tag{5.1}$$

where $c_\nu = \binom{d}{\nu} \gamma(n+d-\nu)$. Turán's lemma (Theorem 2.4) applied on the right-hand side of (5.1) shows that $P^{d,n}(x)$ is hyperbolic if

$$\sum_{\nu=0}^{d-2} 2^\nu \nu! c_\nu^2 < 2^d (d-1)! c_d^2 = 2^d (d-1)! \gamma(n)^2.$$

However,

$$\frac{2^\nu \nu! c_\nu^2}{2^d (d-1)! \gamma(n)^2} = \frac{d}{2^{d-\nu} (d-\nu)!} \binom{d}{d-\nu} \frac{\gamma(n+d-\nu)^2}{\gamma(n)^2}$$

and we are done. □

Stirling's formula implies that

$$\frac{n!}{(2n)!} \sim \frac{\sqrt{2\pi} e^{-n} n^{n+1/2}}{\sqrt{2\pi} e^{-2n} (2n)^{2n+1/2}} = \frac{1}{\sqrt{2}} \cdot \left(\frac{e}{4n} \right)^n.$$

This combined with the first term of the asymptotic expansion for $\xi^{(2n)}(1/2)$ (i.e. Theorem 4.2 with $K = 1$) proves

Proposition 5.3. *As $n \rightarrow \infty$,*

$$\gamma(n) = \Psi(n) \left(1 + O\left(\frac{\log(n)}{n} \right) \right)$$

where

$$\Psi(n) = 4\pi^2 e^{7w/4} \sqrt{\frac{w}{w+1}} \left(\frac{ew^2}{16ne^{2/w}} \right)^n$$

and $w = W(2n/\pi)$.

Proposition 5.4. *Suppose $c \geq 1$. For n sufficiently large and $0 \leq j \leq n^c - n$, we have*

$$\frac{\gamma(n+j)}{\gamma(n)} \leq 1.01 \left(\frac{3c^2 \log(n)^2}{16n} \right)^j.$$

Proof. Proposition 5.3 implies, for any $\delta > 0$, there exists an $N > 0$ such that

$$\Psi(n)(1 - \delta) \leq \gamma(n) \leq \Psi(n)(1 + \delta)$$

for all $n \geq N$. For sufficiently small $\varepsilon > 0$, if we take $\delta = \varepsilon/3$, then

$$\frac{\gamma(n+j)}{\gamma(n)} \leq \frac{\Psi(n+j)}{\Psi(n)}(1 + \varepsilon) \quad (5.2)$$

for all $n \geq N$ and all $j \geq 0$. We can write, on rearranging,

$$\frac{\Psi(n+j)}{\Psi(n)} = \sqrt{\frac{1+1/w}{1+1/w_j}} \left(\frac{w_j^2}{w^2} \cdot \frac{n}{n+j} \right)^n \exp \left(\frac{7(w_j - w)}{4} + \frac{2n(w_j - w)}{ww_j} \right) \left(\frac{ew_j^2}{16(n+j)e^{2/w_j}} \right)^j$$

where $w_j = W(2(n+j)/\pi)$. We will establish bounds for the term of (5.2) as follows. Using (4.1) with $c = 1 + j/n$, we get

$$w_j - w = W(c \cdot 2n/\pi) - W(2n/\pi) \leq \log(1 + j/n). \quad (5.3)$$

Thus,

$$\begin{aligned} \left(\frac{w_j}{w} \right)^{2n} &\leq \left(\frac{w + \log(1 + j/n)}{w} \right)^{2n} \\ &= \left(1 + \frac{\log(1 + j/n)}{w} \right)^{w \cdot 2n/w} \\ &\leq e^{\log(1+j/n) \cdot 2n/w} \leq e^{2j/w} \end{aligned}$$

where we used the fact that

$$\left(1 + \frac{a}{b} \right)^b \leq e^a$$

for $a \geq 0$ and $b > 0$. Using (5.3) again and the fact that W is an increasing function,

$$\begin{aligned} \exp \left(\frac{7(w_j - w)}{4} + \frac{2n(w_j - w)}{ww_j} \right) &\leq \exp \left(\frac{7 \log(1 + j/n)}{4} + \frac{2n \log(1 + j/n)}{w^2} \right) \\ &\leq \exp \left(\frac{7j}{4n} + \frac{2j}{w^2} \right). \end{aligned}$$

Since $1/(1 + 1/w_j)$ and $n/(n+j)$ are both less than 1, we have

$$\sqrt{\frac{1+1/w}{1+1/w_j}} \left(\frac{n}{n+j} \right)^n \leq \sqrt{1 + \frac{1}{w}}.$$

We pick n large enough such that $(1 + \varepsilon)\sqrt{1 + 1/w} \leq 1.01$. Then (5.2) gives us

$$\frac{\gamma(n+j)}{\gamma(n)} \leq 1.01 \left(\frac{w_j^2}{16(n+j)} \exp \left[\frac{7}{4n} + \frac{2}{w} + \frac{2}{w^2} - \frac{2}{w_j} + 1 \right] \right)^j.$$

Since $W(x) \rightarrow \infty$ as $x \rightarrow \infty$, we can make the exponential term in the bound above close to $e < 3$ for n large enough. For such n ,

$$\frac{\gamma(n+j)}{\gamma(n)} \leq 1.01 \left(\frac{3w_j^2}{16(n+j)} \right)^j \leq 1.01 \left(\frac{3 \log(2n^c/\pi)^2}{16n} \right)^j < 1.01 \left(\frac{\log(n^c)^2}{16n} \right)^j.$$

where we used the fact that $W(x) \leq \log(x)$ and $c \geq 1$. $\square$

Theorem 5.2 (O'Sullivan). *For all d sufficiently large, $P^{d,n}(x)$ is hyperbolic whenever*

$$\frac{n}{\log^2(n)} \geq \frac{1}{2}d^{3/4}. \quad (5.4)$$

Proof. Taking $c = 2$ in Proposition 5.4, we have

$$\frac{\gamma(n+j)^2}{\gamma(n)^2} < 1.03 \left(\frac{9 \log(n)^4}{16n^2} \right)^j$$

since $1.01^2 = 1.0201 < 1.3$. Then, with Proposition 5.2 in mind, we see that

$$\sum_{j=2}^d \frac{d}{2^j j!} \binom{d}{j} \frac{\gamma(n+j)^2}{\gamma(n)^2} < 1.03 \sum_{j=2}^d \frac{d}{(j!)^2} \left(\frac{9d \log(n)^4}{32n^2} \right)^2$$

where we used the fact that $\binom{d}{j} < \frac{d!}{j!}$. We choose n large enough so that

$$\left(\frac{9d \log(n)^4}{32n^2} \right)^2 \leq \frac{3}{d}. \quad (5.5)$$

This is possible if and only if

$$\frac{n}{\log^2(n)} \geq \frac{3^{3/4}}{2^{5/2}} d^{3/4} \approx 0.403 d^{3/4}.$$

In particular, inequality (5.5) holds for all n that satisfy (5.4). Then

$$\sum_{j=2}^d \frac{d}{2^j j!} \binom{d}{j} \frac{\gamma(n+j)^2}{\gamma(n)^2} < 1.03 \sum_{j=2}^d \frac{d}{(j!)^2} \left(\sqrt{\frac{3}{d}} \right)^j < 1.03 \sum_{j=2}^{\infty} \frac{3^{j/2}}{(j!)^2} < 0.94 < 1$$

which satisfies the hypothesis of Proposition 5.2. $\square$

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